DGA: Direction-Guided Attack Against Optical Aerial Detection in Camera Shooting Direction-Agnostic Scenarios

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***Abstract*— Patch-based adversarial attacks have increasingly aroused concerns due to their application potential in military and civilian fields. In aerial imagery, numerous targets exhibit inherent directionality, such as vehicles and ships, giving rise to the emergence of oriented object detection tasks; similarly, adversarial patches also exhibit intrinsic orientation due to their lack of perfect symmetry. Existing methods presuppose a static alignment between the adversarial patch's orientation and the camera's coordinate system—an assumption that is frequently violated in aerial images, whose effectiveness degrades in real-world scenarios. In this article, we investigate the often-neglected aspect of patch orientation in adversarial attacks and its impact on camouflage effectiveness, particularly when the orientation is not congruent with the target. A new direction-guided attack (DGA) framework is proposed for deceiving real-world aerial detectors, which shows robust and adaptable attack performance in camera shooting direction-agnostic (CSDA) scenarios. The core idea of DGA is to utilize affine transformations to constrain the relative orientation of the patch to the target and introduce three types of loss to reduce target detection confidence, make the color printable, and smooth the patch color. We introduce a direction-guided evaluation methodology to bridge the gap between patch performance in the digital domain and its actual real-world efficacy. Moreover, we establish a drone-based vehicle detection dataset (SJTU-4K), which labels the orientation of the target, to assess the robustness of patches under various shooting altitudes and views. Extensive proportionally scaled and 1:1 experiments are performed in physical scenarios, demonstrating the superiority and potential of the proposed framework for real-world attacks.**

***Index Terms*— Adversarial attack, aerial detection, camouflage, physical attack.**

I. Introduction

DEEP neural networks (DNNs) have achieved remarkable success in remote sensing images (RSIs), significantly accelerating the application of artificial intelligence (AI) in remote sensing. However, some works [1], [2] have demonstrated that DNN-based models exhibit a lack of stability about specially designed "small" perturbations to the signal input space [3]. Images containing such disturbances are usually referred to as adversarial examples [4]. These "small" disturbances keep the signal content perceived by humans unchanged while causing fundamental changes in the output of the DNN-based model, which causes potential security risks. With people's attention to AI security, research on adversarial attacks is gradually gaining attention [5], [6], [7]. The development speed of AI has lost control. We believe that learning how to attack AI now may be learning how to protect ourselves in the future. By studying adversarial samples, people can better understand the black box of DNN models and construct more robust DNN algorithms. Attack and defense are double-edged swords, and only by a better understanding of potential risks can we better defend against them.

Nowadays, aerial detection has played a vital role in military and civilian fields [8], [9], [10]. In the past, people often used camouflage or camouflage nets to disguise important ground targets, making it impossible for the naked eye to detect them from aerial images [11]. However, with the widespread application of deep learning, these traditional disguises are meaningless in front of powerful DNN aerial detectors [12], [13]. Fortunately, aerial detectors are also fragile and can be easily fooled by adversarial samples [14], [15]. Nonetheless, most existing attacks are designed for digital-level attacks, which create subtle perturbations by modifying pixels in the image [16]. However, pixel-level disturbances are difficult to capture by optical cameras on aircraft and therefore cannot be applied to the real world. Thus, some work began to attempt physical-level attacks by placing malicious patches on the target object, making the target undetectable. When the size of the adversarial patch is close to the target object, it can be captured by the optical camera on the aircraft and implanted for attack. The existing patch attack methods [14], [17], [18], [19], [20] mostly originate directly from general real-scenario attack methods [21], [22]. Natural images often imply the assumption that the sky is above and the ground is below, so patch attacks do not need to consider orientation issues. However, the direction of the target in aerial images changes, so the orientation of the patch pasted on the target should change along with the target, as shown in Fig. 1. Most existing methodologies consistently align the orientation of patches with the camera coordinate system, which is incongruent with real-world scenarios. In addition, there have been recent attempts to install patches in the background

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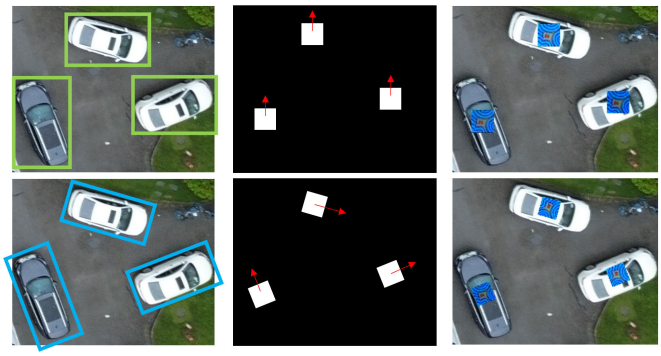


Fig. 1. Difference between regular patch attack and directional patch attack.

around the target [23], [24], but these approaches demand that targets remain at predetermined locations, resulting in a lack of operational flexibility.

To solve the above problems, we propose an innovative physical attack approach called direction-guided attack (DGA), which can protect targeted objects from recognition in scenes where the camera's shooting direction is arbitrary. Specifically, patches are generated under the guidance of the target's orientation information. During the training process, the direction of the patch is ensured to be consistent with the target orientation, thereby enabling the patch to achieve robustness to the camera shooting direction. Besides, we have designed a new patch-based attack evaluation method to reduce the gap between digital and real-world evaluation. We created a vehicle aerial detection dataset using a drone and conducted extensive experiments at the digital, physical, and real-world levels to verify the effectiveness of the proposed DGA and evaluate the robustness of adversarial attacks against various aerial detectors.

The main contributions of this work are summarized as follows.

1) A novel DGA framework is devised to deceive aerial detection methods in the real world, which can gift the patches with state-of-the-art (SOTA) attack performance in both white- and black-box settings in camera shooting direction-agnostic (CSDA) scenarios.
2) A brand-new patch-based attack evaluation method for aerial images has been proposed, eliminating the inconsistency between the patch-pasting method in digital and the real world. Moreover, we comprehensively benchmark adversarial patches against the 15 aerial detectors under this new evaluation method.
3) We collect a lot of vehicle data using drones on our campus and create an aerial vehicle detection dataset, called SJTU-4K. It provides directional information about the car to guide the patch-pasting position and aerial images captured at different altitudes, pitch angles, and directions, which are closer to real scenes.
4) Extensive proportionally scaled experiments and 1:1 experiments are performed in physical scenarios, demonstrating the superiority and potential of the proposed framework for real-world attacks. We further discuss the impact of factors such as shooting altitude, pitch angle, patch position, lighting condition, and the material on the attack efficacy, which is expected to have reference value for future research.

The rest of this article is organized as follows. The related works are introduced in Section II. In Section III, the details of the proposed DGA framework are introduced. Then, we verify the effectiveness of the proposed DGA at the digital, physical, and real-world levels and demonstrate the advantages of the generated direction-guided patches in Section IV. Next, we further discuss the imperceptibility of patch attacks in Section V. Finally, conclusions are presented in Section VI.

II. Related Work

Patch-based attacks against optical aerial detection can be classified into four categories based on the positional relationship between the patch and target, as shown in Table I. This section provides a detailed explanation of them.

A. Patch on the Target

This method is the most direct patch attack, which just puts patches on the target to evade detection by aerial detectors [14]. Since the size of the target is small in aerial images, the location and size of patch pasting are very limited. Patch on the target not only has lower printing costs but also is easy to install. Besides, the shape and size of the patch are standardized and suitable for mass production. The DGA proposed in this article belongs to this type of patch-based attack method.

B. Patch Outside the Target

Den Hollander et al. [17] show that the larger the patch, the better the attack effect. There are usually very few areas above the target where patches can be placed, which limits the size of the patches. Therefore, Lian et al. [23] propose to place patches outside the target and improve attack performance by increasing the patch area. However, due to the large patch area, it increases its installation difficulty and printing cost. Meanwhile, due to the need for special printing equipment for large patches, their standardization and mass production capabilities will be reduced. Another drawback of this method is that it can only attack targets parked in specific areas.

C. Patch Under the Target

Lian et al. [25] found that the closer the patch is to the target, the better the attack effect is, so they attempted to place the patch under the target. This can make the patch closest to the target and also ensure that the size of the patch is not limited by the size of the target. However, since the patch is pasted on the background rather than the target, this means that the attack necessitates that the targets remain at predetermined locations, which lacks flexibility.

D. 3-D Patch Covering the Target

Compared to the three 2-D patches mentioned earlier, 3-D patches can better cover targets and prevent monitoring from

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TABLE I
MAINSTREAM RESEARCH DIRECTIONS FOR PHYSICAL PATCHES IN RECENT YEARS

Method	Parking position	Printing costs	Installation difficulty	Standardization	Mass production
Patch on the target	Flexible	Low	Easy	High	Easy
Patch outside the target	Fixed	Medium	Medium	Medium	Medium
Patch under the target	Fixed	Medium	Medium	Medium	Medium
3D patch covering the target	Flexible	High	Hard	Low	Hard

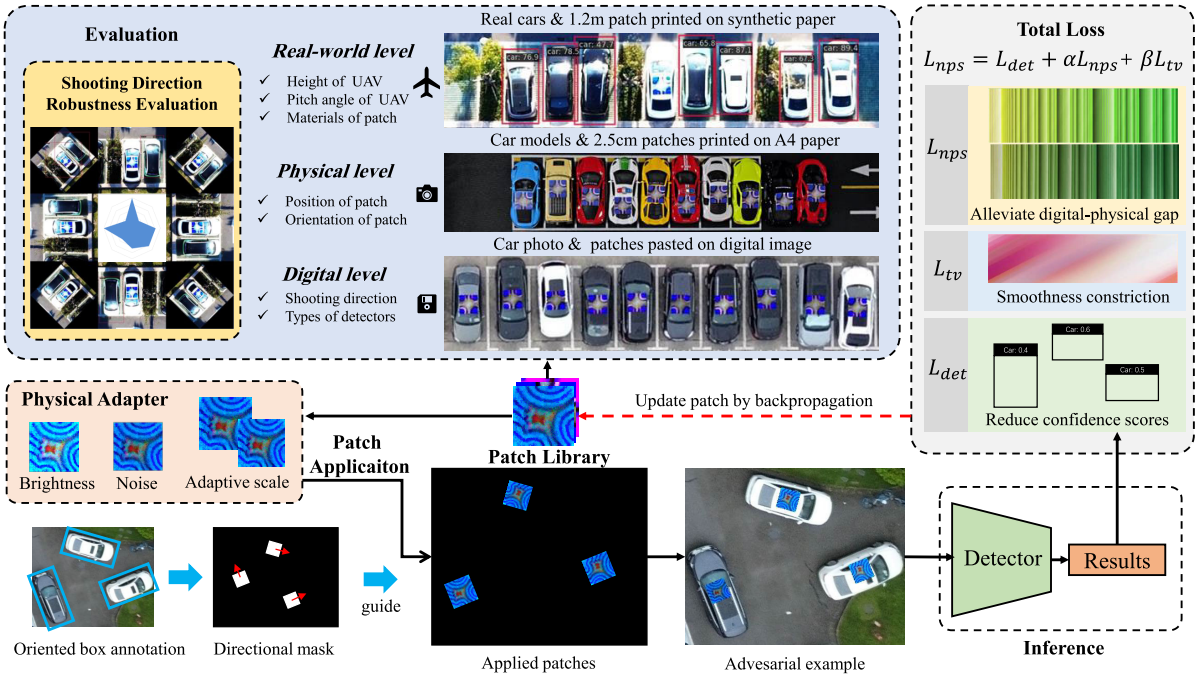


Fig. 2. Illustration of the proposed directional patch attack method.

various angles [26]. It imposes no requirements on the docking locations of the targets. However, the process of producing 3-D patches is more complex, resulting in higher printing costs and installation difficulties. Besides, since the appearance of different targets is not exactly the same, 3-D patches for different targets need to be customized. This means that the standardization level of 3-D patches is low, making it difficult to carry out large-scale production.

III. METHODOLOGY

In this section, we formulate the problem and then detail the proposed DGA with vehicles as the targets of interest. It is believed that DGA can also work for other interested targets similarly. The overall pipeline is shown in Fig. 2.

A. Problem Formulation

The purpose of aerial detection is to detect interested targets from aerial images, while the purpose of adversarial attacks is to prevent aerial detectors from discovering these targets. This means that adversarial attacks can be used to conceal and protect important assets. In the real world, the transformation of an object's model coordinates to the final 2-D coordinates displayed on a digital screen involves model-view-projection (MVP) transformations [27].

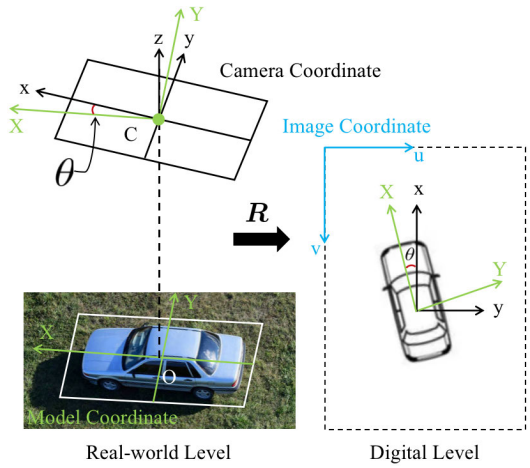


Fig. 3. Transformation process of the target from model coordinates to camera coordinates.

Fig. 3 provides a schematic of the MVP process. We assume that the origin of the model coordinate system is at point O, with the X-axis and Y-axis parallel to the ground plane. The origin of the camera coordinate system corresponds to the camera's optical center C, with the x-axis and y-axis also parallel to the ground plane. The angle between the camera coordinate system's x-axis and the vehicle's heading

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direction determines the vehicle's orientation in the image. Considering the omission of environmental factors such as weather and lighting conditions, the process can be denoted by the transformation matrix $\boldsymbol{R}$, which is expressed as

$$\boldsymbol{R} = \boldsymbol{M}_i \boldsymbol{M}_e \tag{1}$$

where $\boldsymbol{M}_i$ denotes the intrinsic matrix of camera and $\boldsymbol{M}_e$ denotes the extrinsic matrix of camera. The intrinsic matrix is generally invariant, while the extrinsic matrix changes with the variations in the drone's attitude.

Assuming that the distance from the image plane to the optical center C of the camera is f, the intrinsic matrix $\boldsymbol{M}_i$ can be represented as

$$\boldsymbol{M}_i = \begin{bmatrix} \frac{f}{dx} & 0 & u_0 & 0 \\ 0 & \frac{f}{dy} & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{2}$$

where dx and dy represent the number of millimeters per pixel in each column and row, respectively; and u_0 and v_0 represent the offset from the origin to the top-left corner of the image.

Assuming that the camera coordinate system rotates by φ, α, and θ degrees around the x-, y-, and z-axes respectively, the extrinsic matrix M_e can be represented in (3), as shown at the bottom of the next page, where t_x, t_y, and t_z represent the offsets of the optical center C relative to the origin of the model coordinate system along the x-, y-, and z-axes, respectively. Please refer to the Appendix for the complete derivation process of $\boldsymbol{M}_i$ and $\boldsymbol{M}_e$.

When the camera captures the images perpendicularly to the ground, the rotation angles of the camera's x-axis and y-axis are both 0. Consequently, $\boldsymbol{R}$ can be further simplified to

$$\boldsymbol{R} = \begin{bmatrix} \frac{f\cos\theta}{dx} & -\frac{f\sin\theta}{dx} & u_0 & \frac{ft_x}{dx} + u_0 t_z \\ \frac{f\sin\theta}{dy} & \frac{f\cos\theta}{dy} & v_0 & \frac{ft_y}{dy} + v_0 t_z \\ 0 & 0 & 1 & t_z \end{bmatrix} \tag{4}$$

where θ represents the angle of rotation about the z-axis of the camera coordinate system, as illustrated in Fig. 3. Let us consider a scenario where the target's real-world orientation is aligned with true north, and the flight path of the drone is charted at an azimuth of 30° relative to the north (bearing east). Under these circumstances, the projected orientation angle, θ, of the target within the image plane would be quantified as 30°.

To preserve the directional information of the target, we adopt a 360° rotated box representation method. It can be obtained through Algorithm 1. The points used for annotation are an ordered set. Point (x_1, y_1) is located on the left front of the target, and point (x_2, y_2) is located on the right front of the target. The line connecting them can accurately indicate the direction of the target's front. Similarly, point (x_3, y_3) is located on the right rear of the target, and point (x_4, y_4) is located on the left rear of the target. The points $(x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4)$ can form a rotated box. The angular range of the rotated box output by the OpenCV function is

Algorithm 1 Generate 360° Rotated Box

Input: points $(x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4)$
Output: rotated box (x_c, y_c, w, h, θ)
1: $\theta \leftarrow$ np.arctan2$(y2 - y1, x2 - x1)$
2: pts $\leftarrow$ points.reshape([-1, 2])
3: $(x_c, y_c), (w, h), \theta_{oc} \leftarrow$cv2.minAreaRect(pts)
4: $\theta = \theta \mod 2\pi$
5: $\theta = \theta - 2\pi$ if $\theta >= \pi$ else θ
6: **return** (x_c, y_c, w, h, θ)

confined within $[-90°, 0°)$, which is hereafter referred to as θ_{oc}. Within the context of this rotated box schema, the width w is aligned with the edge that constitutes an acute angle with the positive direction of the x-axis, while the height h corresponds to the orthogonal edge. Contrary to the direct application of this convention, the present study adopts an alternative approach for angle determination. The angle θ is computed utilizing the arctangent function, applied to an ordered set of points, thereby ensuring that θ falls within the range $[-\pi, \pi)$.

Given a target's clean image $\boldsymbol{X}$, the attacked image $\boldsymbol{X}^\dagger$ can be represented as

$$\boldsymbol{X}^\dagger = (1 - \boldsymbol{M_P}) \odot \boldsymbol{X} + \boldsymbol{M_P} \odot \boldsymbol{P} \tag{5}$$

where $\odot$ is the Hadamard product and $\boldsymbol{P}$ and $\boldsymbol{M_P}$ denote the adversarial patch and the mask of patch, respectively. $\boldsymbol{M_P}$ is a binary matrix, with a value of 1 for the patch position area and 0 for the other areas.

B. Assumption of Patch Coordinate System

As targets have their model coordinate system, patches pasted onto targets also possess their coordinate system. Previous adversarial attacks against aerial detection usually have the following assumption.

Assumption 1: The patch coordinate system is always aligned with the camera coordinate system.

It implies that the orientation of the patch is always directed toward the negative half-axis of the x-axis in the image coordinate system, as shown in the first row of Fig. 1. This assumption first emerged in patch attacks against ground images [21] and has been directly applied to patch attacks against aerial detectors [14], [15]. Due to the influence of gravity, the camera coordinate system in ground images typically assumes the x-axis to be perpendicular to the ground and the y-axis to be parallel to the ground; thus, Assumption 1 is generally satisfied in most cases.

One of the differences between aerial imagery and ground imagery is that some targets in aerial images have an inherent orientation attribute, which is referred to as oriented targets. As shown in Fig. 3, the orientation of these targets varies with changes in the camera coordinate system. After a patch is printed and affixed to the surface of a target, its orientation also changes with the camera coordinate system. This means that Assumption 1 is violated. Addressing the unique scenario of unknown camera shooting angles in aerial imagery, we propose a new assumption instead of Assumption 1.

Assumption 2: The patch coordinate system is always aligned with the target coordinate system.

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This assumption aligns more closely with the practical applications of patch attacks, acknowledging the inherent orientation attributes of targets and patches in aerial imagery. The advantages of the new assumption are twofold.

1) *Training Alignment:* Previous methods always fixed the orientation of the patch during the training phase to match the camera's shooting direction. This becomes problematic when the camera's shooting direction changes, which is common in real-world scenarios, leading to a reduction of the patch's attack performance. With the new assumption, the orientation of the patch remains consistent with the direction of the target regardless of changes in the camera's shooting direction. As a result, the patches trained under this assumption exhibit robustness to scenarios where the camera's shooting angle is unknown.
2) *Testing Alignment:* As for previous methods, the orientation of the patch does not change in response to target movements during the testing phase. However, in practical camouflage scenarios, since targets often have rigid bodies, the orientation of patches pasted onto them will change as the orientation of the targets changes. Therefore, the digital-level evaluation of patches based on the new assumption yields results that are more aligned with the real-world scenario.

C. Directional Patch Adaption and Application

Since the DGA proposed in this article aims to generate adversarial patches with solid attack efficiency in real-world scenarios, we applied a series of image transformations simulating the real dynamic physical environment during the patch adaption phase, including varying brightness, different noises, and adaptive scales. In previous work on natural images [22], to prevent camera shaking during the shooting process, random rotation is usually added in the patch adaptation to increase the robustness of the patch. The default range of random rotation is set between $-20°$ and $20°$. However, for aerial images, the angle variation range of the target is relatively large, ranging from $-180°$ to $180°$. It is obviously unreasonable to directly apply the previous configuration to aerial detection. In the DGA training phase, we have removed the random rotation, as we will make more precise adjustments to the direction of the patch during the patch application phase. The above physical adaptation can be represented by the patch conversion function PT

$$\boldsymbol{P}^t = \mathrm{PT}(\boldsymbol{P}) \tag{6}$$

where $\boldsymbol{P}^t$ denotes the augmented patches, which are shown in the physical adapter of Fig. 2. PT is a data augmentation function that applies random contrast, brightness, and noise to the patch $\boldsymbol{P}$, which can be expressed as

$$\boldsymbol{P}^t_{i,j} = (\boldsymbol{P}_{i,j} \cdot c) + b + n$$

where $\boldsymbol{P}_{i,j}$ represents the pixel value in the ith row and jth column of the patch. c is a random contrast factor and b is a random brightness adjustment, both of which are typically defined as $c \sim \mathcal{U}(c_{\min}, c_{\max})$ and $b \sim \mathcal{U}(b_{\min}, b_{\max})$, sampled from uniform distributions with their respective minimum and maximum bounds. The term n is random noise added to the image, defined as $n \sim \mathcal{U}(n_{\min}, n_{\max})$, which is sampled from a uniform distribution between $n_{\min}$ and $n_{\max}$. In Section IV, the values of $c_{\min}$, $c_{\max}$, $b_{\min}$, $b_{\max}$, $n_{\min}$, and $n_{\max}$ are 0.8, 1.2, -0.1, 0.1, -0.1, and 0.1, respectively.

Next, the elaborated adversarial patches must be pasted on the clean images with proper size and position. This process is called patch application. Usually, during the training process, patches should be placed where they work in the actual application. This location not only needs to include the coordinates of the patch but also needs to provide the orientation information of the patch. Therefore, we attempt to align the direction of the patch with the direction of the car head through quantitative rotation, as shown in Fig. 2. In practical applications, we can use a rotated box detector to predict the angle information of vehicles [28]. Nowadays, research on rotated box detectors is mature, and the prediction of vehicle orientation is accurate.

The process of directional patch application is shown in Fig. 4. First, we apply zero padding to the edges of the adaptively enhanced patch to match the dimensions of image X. Assume that the dimensions of the patch are $w_p \times h_p$ and the dimensions of the image are $W \times H$. Specifically, we pad the top and bottom of the patch with $(H - h_p)/2$ pixels and the left and right sides with $(W - w_p)/2$ pixels. The output of the rotated box detector can be represented as $\boldsymbol{y} = (x_c, y_c, w, h, \theta)$. The patch is then moved to the specified location via an affine transformation. To facilitate description, this affine transformation can be decomposed into three discrete operations, i.e., translation, rotation, and scaling. A 3×3 homogeneous coordinate matrix denotes the affine transformation on a 2-D plane. Initially, the padded patch is translated from the center point (W, H) to the origin $(0, 0)$ and subsequently moved from the origin to the target's center point (x_c, y_c). By multiplying these two transformation matrices, the translation matrix is presented as follows:

$$\begin{aligned} J_{\mathrm{tra}} &= \begin{bmatrix} 1 & 0 & -W/2 \\ 0 & 1 & -H/2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & x_c \\ 0 & 1 & y_c \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & x_c - W/2 \\ 0 & 1 & y_c - H/2 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned} \tag{7}$$

$$\boldsymbol{M}_e = \begin{bmatrix} \cos\theta\cos\alpha - \sin\theta\sin\varphi\sin\alpha & -\sin\theta\cos\varphi & -\cos\theta\sin\alpha - \sin\theta\sin\varphi\cos\alpha & t_x \\ \sin\theta\cos\alpha + \cos\theta\sin\varphi\sin\alpha & \cos\theta\cos\varphi & -\sin\theta\sin\alpha + \cos\theta\sin\varphi\cos\alpha & t_y \\ \cos\varphi\sin\alpha & -\sin\varphi & \cos\varphi\cos\alpha & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

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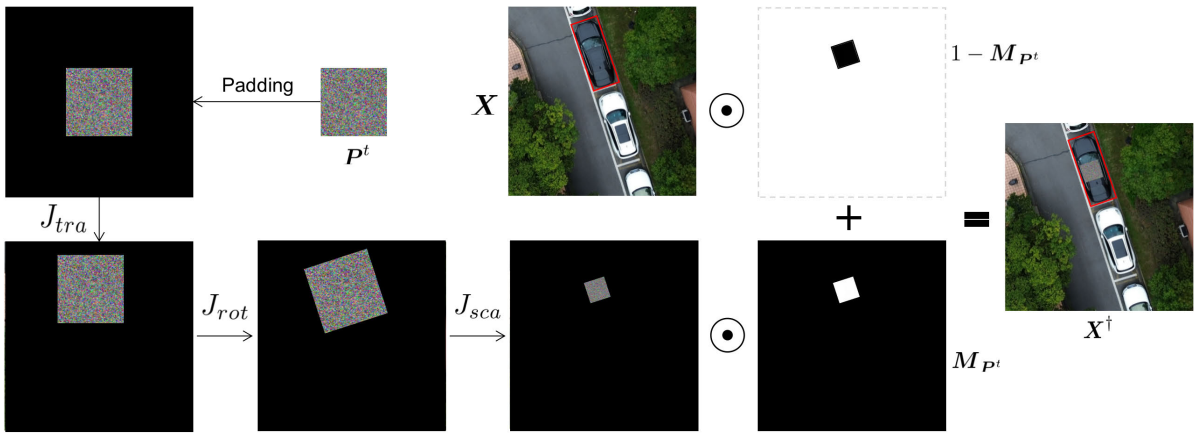


Fig. 4. Detailed process of directional patch application.

Next, the patch is rotated by an angle corresponding to the target's orientation information. The rotate matrix can be written as

$$J_{\text{rot}} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{8}$$

Subsequently, the scaling ratio S is calculated based on the width and height of the target to the initial size of the patch as follows:

$$S = \gamma \frac{\sqrt[2]{w^2 + h^2}}{w_p} \tag{9}$$

where w and h represent the width and height of the rotated bounding box, respectively. γ is the coefficient for adaptively adjusting the patch size, with a default setting of 0.2. It can limit patches not to exceed the boundaries of the target. Thus, the scaling matrix can be written as

$$J_{\text{sca}} = \begin{bmatrix} S & 0 & 0 \\ 0 & S & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{10}$$

Next, by multiplying these three matrices, the entire affine transformation can be represented using a single transformation matrix as follows:

$$\begin{aligned} J &= J_{\text{tra}} \cdot J_{\text{rot}} \cdot J_{\text{sca}} \\ &= \begin{bmatrix} S\cos\theta & -\sin\theta & x_c - W/2 \\ \sin\theta & S\cos\theta & y_c - H/2 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned} \tag{11}$$

Finally, the binary mask of adversarial patch $\boldsymbol{M}_{\boldsymbol{P}^t}$ is formulated as

$$\boldsymbol{M}_{\boldsymbol{P}^t} = \begin{cases} 1, & \text{if } p_{i,j} \neq 0 \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $0 \leq i \leq W-1, 0 \leq j \leq H-1$. $p_{i,j}$ is the pixel value of the patch after affine transformation. Finally, the originally formulated problem (2) can be transformed as

$$\boldsymbol{X}^\dagger = (1 - \boldsymbol{M}_{\boldsymbol{P}^t}) \odot \boldsymbol{X} + \boldsymbol{M}_{\boldsymbol{P}^t} \odot F_{\text{affine}}(\boldsymbol{P}^t, J) \tag{13}$$

where F_{affine} is an affine transformation function that can apply an affine transformation represented by J to the patch.

D. Loss Design

In this article, we aim to perform physical attacks, i.e., to hide the specified targets from being detected by optical aerial detectors in the real world. Therefore, the objectiveness function comprises adversarial confidence, nonprintable score, and smoothness constriction loss.

To prevent the aerial detector from detecting the specified targets, it is essentially necessary to reduce the detector's confidence in the predicted results. Therefore, we design an adversarial confidence loss, which is written as

$$L_{\text{conf}} = \frac{1}{n} \sum_{i=1}^{n} C_i \tag{14}$$

where n is the total number of predicted objects and C_i denotes the confidence score of the ith predicted object.

Patch attacks require printing on adversarial media to affect the physical world, and the colors that printers can print are limited. Although some colors exhibit advantages in digital-level attacks, they cannot be printed. Hence, nonprintable score [29] is adopted to alleviate the digital-physical gap, which can be written as

$$L_{\text{nps}} = \sum_{m,n} \min_{\boldsymbol{c}_{\text{print}} \in \boldsymbol{C}} \sqrt{(\boldsymbol{P}_{m,n} - \boldsymbol{c}_{\text{print}})^2 + \xi} \tag{15}$$

where $\boldsymbol{P}^t_{m,n}$ means the pixel value in the ith row and jth column of the patch, $\boldsymbol{C}$ denotes the printable color set, and $\boldsymbol{c}_{\text{print}}$ is one color of $\boldsymbol{C}$. The smaller L_{nps}, the smaller the color difference between the printed physical-level patch and the digital-level patch. ξ is a minimum value used to ensure that the calculation of the square root is differentiable, and it is usually set to $1e^{-6}$.

For the physical attack, it is hard for detection systems to capture the gap between adjacent pixels. Hence, total variation [29] is adopted to constraint the smoothness of the generated adversarial patch, which can be written as

$$L_{\text{tv}} = \sum_{m,n} \sqrt{\left(\boldsymbol{P}^t_{m+1,n} - \boldsymbol{P}^t_{m,n}\right)^2 + \left(\boldsymbol{P}^t_{m,n+1} - \boldsymbol{P}^t_{m,n}\right)^2}. \tag{16}$$

The smaller L_{tv}, the smoother the color change of the patch.

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Finally, the overall objective loss is as follows:

$$L = L_{\text{conf}} + \alpha L_{\text{nps}} + \beta L_{\text{tv}} \tag{17}$$

where α and β are hyperparameters to scale L_{nps} and L_{tv}, respectively. Our proposed DGA aims to minimize the objective function L and optimize the adversarial patch, so we freeze all the weights and biases in the aerial detection model and only update the pixel values of the adversarial patch.

E. Overall Training Procedures

The pipeline of the proposed DGA patch-based attack method is displayed in Fig. 2. To facilitate understanding and reproduction, we have provided a detailed description of the training process of DGA in Algorithm 2. Specifically, given the original patch $\boldsymbol{P}^0$, after a series of physical, scale adaptive, and rotation transformations, patch $\boldsymbol{P}^t$ will be placed on a clean image X to form an adversarial image X. Next, adversarial examples will be input into the target aerial detector. Then, the confidence score extracted from the detection results can be used as part of the total loss. The next step is to backpropagate based on the total loss L. Adversarial patch $\boldsymbol{P}$ will be updated. Finally, repeat the above steps until the preset number of training epochs is reached.

Algorithm 2 DGA

Input: Detector $D(\cdot)$, an clean image $\boldsymbol{X}$, ground truth $\boldsymbol{y} = (x_c, y_c, w, h, \theta)$, initial patch $\boldsymbol{P}^0$, total objective loss $L(\cdot)$, the number of training epochs N_{epoch}, the number of training set images N_{imgs}, hyperparameters α and β.

Output: an adversarial patch $\boldsymbol{P}$

Initialize $\boldsymbol{P}^0$ randomly in [0, 255], $\boldsymbol{P} \leftarrow \boldsymbol{P}^0$
for $i = 0$ to N_{epoch} **do**
  for $j = 0$ to N_{imgs} **do**
    $\boldsymbol{P}^t = F_{\text{robust}}(\boldsymbol{P})$
    $\boldsymbol{X}^{\dagger} = \left(1 - \boldsymbol{M}_{\boldsymbol{P}}^t\right) \odot \boldsymbol{X} + \boldsymbol{M}_{\boldsymbol{P}^t} \odot F_{\text{affine}}(\boldsymbol{P}^t, J)$
    $L_{conf} \leftarrow D(\boldsymbol{X}^{\dagger})$
    $L = L_{conf} + \alpha L_{nps} + \beta L_{tv}$
    $\boldsymbol{P}_{i,j+1} = \boldsymbol{P}_{i,j} + \eta \hat{\nabla}_{\boldsymbol{P}_{i,j}} L$
return $\boldsymbol{P} \leftarrow \boldsymbol{P}_{N_{epoch}, N_{imgs}}$

F. Toward Real-World Scenarios Patch Evaluation

In existing patch-based attack evaluation methods, the direction of the patch is independent of the target. For natural images, this is not inappropriate. However, if aerial images also use this evaluation method, it will lead to a gap between the evaluation results and the actual effects in the real world. In the physical world, patches are printed onto adversarial media and then pasted onto the target surface.

Therefore, considering practical application scenarios, to better evaluate the actual efficacy of patches in the physical world, this article proposes a new patch-based attack evaluation method different from previous methods. The difference between the two evaluation methods can be referred to in the two schematics on the right of Fig. 1. The targeted object is often a rigid body, which means that once the patch is pasted on the surface of the target, it will establish a constraint relationship with the target

$$\theta_{\text{target}} = \theta_{\text{patch}}. \tag{18}$$

This constraint relationship ensures the consistency between the direction of the target and the patch in the physical world.

After the patches are pasted, we use the average precision (AP) to evaluate the detection results of these aerial detectors. The conservative 0.45 intersection over union (IoU) and 0.4 confidence thresholds were chosen to meet real-world concerns. Especially for military scenarios, it is reasonable to foresee a military using conservative benchmarks for true positive detection as a partially correct detection still poses a risk to assets or human life [30]. Thus, we will desire a patch capable of suppressing conservative detections.

IV. Experiment and Analysis

In this section, we perform comprehensive experiments to validate the convincingness of the proposed DGA framework. We first outline the SJTU-4K dataset[1] and the experimental settings and then separately report the results that come from the digital, physical, and real-world levels. Subsequently, we conduct a series of exploratory experiments on variables that may affect the experimental results in reality.

A. Dataset

To facilitate adversarial attacks on objects from multiple scales and perspectives, it is essential to have a dataset comprising images captured at various altitudes and camera pitch angles. As shown in Table II, existing datasets of ground vehicles derived from unmanned aerial vehicle (UAV) surveillance suffer from significant shortcomings: they either fail to provide sufficient information on the orientation of targets or lack the necessary diversity in terms of multiple scales and viewpoints. These deficiencies hinder the development and evaluation of adversarial techniques designed for real-world applicability.

In response to this deficiency, this work introduces the SJTU-4k dataset, an expansive collection of UAV-captured vehicle imagery specifically designed to enable comprehensive experimentation with multiscale, multiview adversarial attacks. Naturally, we consider collecting data with a drone from different heights, orientations, and pitch angles. To collect the data we require, we design a reasonable scheme for data capture. First, we chose 20 scenes on our campus as our experimental site, including streets and car parks, where many kinds of cars are often seen. Afterward, we used the DJI Mini 3 drone as the capturing tool. In our scheme, the flight height ranges from 20 to 120 m, and there were nine flight heights in total. The resolution to the raw images is 4000 × 2260 pixels. We also provide aerial images captured at different pitch angles and directions, which are closer to real scenes. Fig. 5 shows several images captured by our drone with different pitch angles from the heights of 20, 40, 80, and 110 m. The size of the objects is quite diverse, and the

[1] https://github.com/zytx121/DGA

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TABLE II
COMPARISON OF UAV-BASED AERIAL VEHICLE DATASETS. NOTE THAT OBB STANDS FOR ORIENTED BOUNDING BOX, AND FRONT INDICATES THE DIRECTION OF THE FRONT OF THE VEHICLE THAT HAS BEEN ANNOTATED

Dataset	#Images	#Instances	Image width	Categories	Altitude	View	OBB	Front
CARPK	1,448	89.8k	1280	1	fixed	fixed		
UAVDT	80,000	841.5k	1100	3	flexible	flexible		
VisDrone	10,209	2600k	2000	10	flexible	flexible		
DroneVehicle	31,064	441.6k	840	5	flexible	flexible	✓	
UAV-ROD	1,577	30.1k	∼1920	1	flexible	fixed	✓	✓
SJTU-4k	16,605	77.1k	1024	1	flexible	flexible	✓	✓

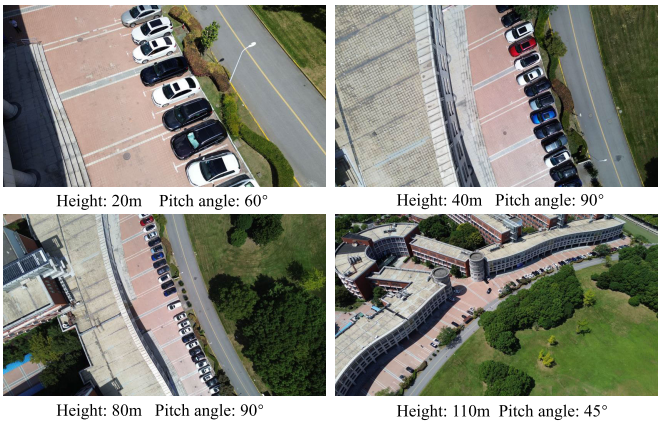


Fig. 5. Multiscale and multipitch angle sample images in the SJTU-4k dataset.

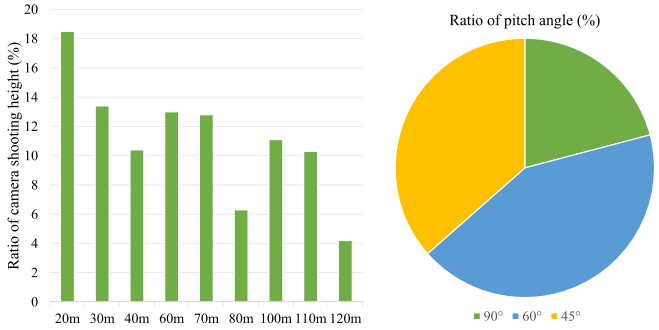


Fig. 6. Distribution of data at different heights and pitch angles in the SJTU-4K dataset.

number of objects in a single image is large, especially for images captured at a great height. The detailed distribution of data in the SJTU-4K dataset is shown in Fig. 6. However, this size is too large, being unsuitable for our experiment. Therefore, we need to conduct some processing of the raw images. They are tailored into a smaller size of 1024 × 1024 when being attacked. Ultimately, there are 10 260 training images with 57 172 car objects and 6345 testing images with 19 969 car objects. To cooperate with the new patch-based attack evaluation method proposed, we used rotated boxes to label vehicles. Unlike the common rotated box representation, we use a 360° angle representation [31] to retain the front information of vehicles. This dataset is a pivotal addition to our research, as it not only furnishes the required target orientation details but also spans a diverse array of viewing angles and scales. Such a rich dataset is critical for simulating and testing adversarial attacks in conditions that closely mimic a variety of operational environments.

B. Experimental Settings

1) Target Models: Dozens of mainstream detectors are adopted to validate the effect of the proposed framework, including YOLOv5-n/s [32], YOLOv6-t/s [33], YOLOv7-t/l [34], YOLOv8-n/s [35], RTMDet-t/s [36], Faster R-CNN [37], Swin Transformer [38], RetinaNet [39], ATSS [40], and RepPoints [41]. Launching attacks on diverse models facilitates the exploration of the susceptibility of varying network architectures and training methodologies to patch-based intrusions. This endeavor not only assists in assessing the robustness of patch attacks but also in gauging their generalizability.

2) Compared Methods: Two SOTA methods, patches on the target, are selected for comparison, including the OBJ [22] and APPA [23]. We adopt the real-world scenarios evaluation method to replace the original evaluation method for all quantitative experiments. APPA-360 is a derivative method of APPA, which adds a random rotation from $-\pi$ to π to the patch during patch adaptation, allowing the patch to obtain the symmetry. It is the most direct idea for dealing with CSDA scenarios. We compare it directly with the directional information-guided DGA proposed in the following experiments.

3) Implementation Details: During training, α and β are set to 0.01 and 2.5, respectively, which is consistent with earlier works on aerial detector attacks [14], [23]. All patches are trained in five periods. This article uses AMSGrad [42] as the optimizer. The learning rate is set to 0.03. All the experiments are carried out on an Ubuntu system with one RTX3090 (24 GB) GPU. Specifically, the YOLO series aerial detectors are developed based on MMYOLO.[2]

Patch-based adversarial attack algorithms require maintaining a parameter count that is directly proportional to the number of patches within a single image during the gradient backpropagation process. In the case of aerial imagery, a single image may contain a large number of targets, each necessitating its patch. Therefore, during the training process, even with a batch size of 1, a substantial amount of GPU memory is utilized. Limited by the GPU memory capacity and the maximum number of targets per image in the dataset, we are constrained to set the batch size to 1.

In physical-level experiments, we use a Xiaomi 13 phone to capture car models. The resolution of the raw images is 3072 × 3072 pixels. We have 20 car models in different colors

[2]https://github.com/open-mmlab/mmyolo

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TABLE III
COMPARISON OF TRAINING TIME OF DIFFERENT METHODS

	Training time
OBJ	0:50:59
APPA	0:54:51
APPA-360	0:56:30
DGA	1:00:02

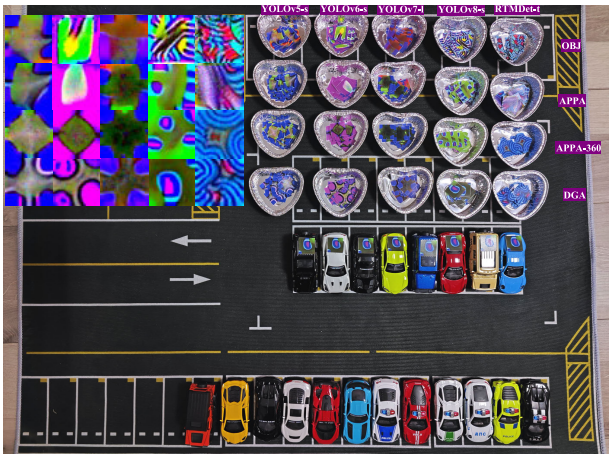


Fig. 7. Experimental materials used for physical-level attack.

TABLE IV
IMPACT OF VARIOUS TRAINING TECHNIQUES ON MODEL ROBUSTNESS

Detector	MixUp	CosineAnnealingLR	AP_{50}
RTMDet-tiny			0.313
	✓		0.820
		✓	0.651
	✓	✓	0.750 (default)
YOLOv8-n			0.085 (default)
	✓		0.410
		✓	0.133
	✓	✓	0.557

and styles, including sedan and off-road vehicles. Their scaling ratio is 1:43, with an average size of 10.5 × 4.5 cm. Adversarial media is regular printing paper. The printer used is an AccurioPrint C3070L. The size of the patches is 2.5 × 2.5 cm. The shooting conditions are indoor, with fluorescent lights on. The shooting background is an 80 × 100 cm carpet with parking patterns. The entire physical-level experimental scenario is shown in Fig. 7. The top-left corner is the digital patches, and the top-right corner is the corresponding printed patches. We printed 20 pieces for each patch and stored them in many small boxes.

In real-world-level experiments, we use a DJI MINI3 drone to capture real cars. The resolution of the raw images is 4000 × 2260 pixels. We used a white BYD Han EV as the target vehicle, with a size of 493 × 186 cm. Adversarial media is synthetic paper, often used to print academic posters. The printer used is an HP Designjet D5800. The size of the patches is 120 × 120 cm. The shooting background is a parking lot with many vehicles.

C. Digital-Level Attack

1) Overall Experimental Results: The direction-guided patches generated by our DGA are shown in Fig. 8. The

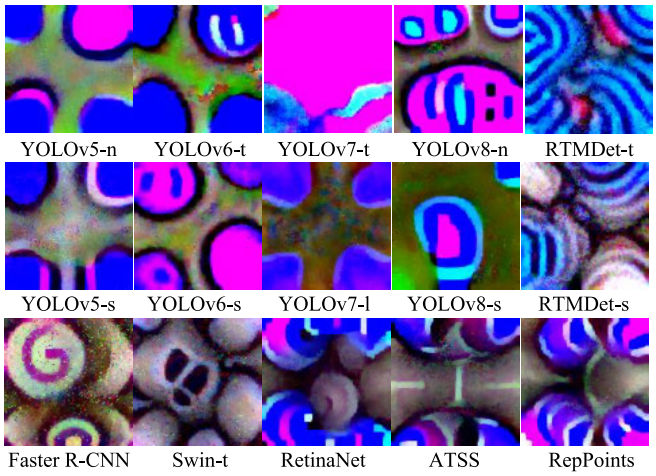


Fig. 8. Adversarial patches generated by the proposed DGA.

patches trained by different aerial detectors have significant differences. It is observed that the following conditions hold.

1) The patches generated by different sizes of YOLOv5 and YOLOv6 are similar, which may be due to their similar network architecture. However, the patches generated by different sizes of YOLOv7 and YOLOv8 are completely different.
2) The patches trained by YOLOv5, YOLOv6, ATSS, and RepPoints tend to have symmetry, but not complete symmetry.
3) Patches trained using single-stage detectors have higher color saturation, while those trained using two-stage detectors have lower color saturation.
4) The patches generated by two-stage CNN-based detectors, represented by Faster R-CNN, and two-stage Transformer-based object detectors, exemplified by Swin Transformer, are significantly different from those produced by other single-stage detectors.

Ten aerial detectors are chosen for the quantitative evaluation. The proposed evaluation method with directional constraints is adopted at the digital level. The experimental results are shown in Fig. 9. Green represents the white-box attacks, black represents the black-box attacks, and bold represents optimal. It is observed that the following conditions hold.

1) For the attackers, DGA achieved the best white-box attack results across ten detectors and ranked first in black-box attacks on nine detectors. Among them, yolov8-n has the strongest black-box attack performance.
2) For the detectors, the Swin Transformer detector has strong robustness against all four patch attacks, but the YOLOV7-l detector is the most vulnerable to patch-based attacks.
3) Patches trained with single-stage detectors exhibit stronger generalization capabilities compared to those trained with two-stage detectors.
4) The generalization capabilities of existing remote sensing patch attack methods are still limited, and they do not achieve satisfactory results on two-stage detectors.

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		D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	Avg.
	Clean	0.936	0.936	0.943	0.941	0.947	0.926	0.936	0.886	0.929	0.947	0.956	0.950	0.930	0.921	0.926	0.934
OBJ	YOLOv5-s (D1)	0.193	0.560	0.238	0.366	0.924	0.313	0.412	0.561	0.789	0.905	0.915	0.901	0.800	0.838	0.830	0.636
	YOLOv6-s (D2)	0.561	0.356	0.384	0.125	0.894	0.469	0.367	0.198	0.623	0.925	0.874	0.896	0.829	0.857	0.837	0.613
	YOLOv7-l (D3)	0.515	0.633	0.312	0.612	0.935	0.531	0.659	0.649	0.839	0.926	0.926	0.916	0.842	0.869	0.872	0.736
	YOLOv8-s (D4)	0.317	0.415	0.216	0.107	0.803	0.440	0.237	0.361	0.554	0.863	0.812	0.831	0.715	0.754	0.655	**0.539**
	RTMDet-t (D5)	0.614	0.711	0.514	0.498	0.761	0.694	0.648	0.377	0.674	0.853	0.813	0.851	0.750	0.764	0.726	**0.683**
	YOLOv5-n (D6)	0.462	0.466	0.290	0.432	0.905	0.322	0.476	0.222	0.646	0.916	0.885	0.888	0.797	0.847	0.804	0.624
	YOLOv6-t (D7)	0.421	0.436	0.350	0.099	0.905	0.396	0.204	0.348	0.618	0.915	0.885	0.903	0.819	0.857	0.827	0.599
	YOLOv7-t (D8)	0.647	0.611	0.507	0.658	0.914	0.578	0.645	0.271	0.705	0.925	0.895	0.891	0.839	0.864	0.838	0.719
	YOLOv8-n (D9)	0.327	0.301	0.231	0.204	0.812	0.345	0.078	0.067	0.166	0.863	0.803	0.659	0.651	0.695	0.513	0.448
	RTMDet-s (D10)	0.457	0.576	0.467	0.472	0.822	0.536	0.373	0.367	0.585	0.813	0.812	0.715	0.675	0.690	0.603	**0.598**
	Faster R-CNN (D11)	0.509	0.671	0.445	0.605	0.823	0.620	0.527	0.546	0.594	0.863	0.803	0.672	0.665	0.614	0.563	**0.635**
	Swin-t (D12)	0.678	0.780	0.636	0.724	0.833	0.706	0.728	0.677	0.673	0.873	0.823	0.789	0.733	0.630	0.659	**0.729**
	RetinaNet (D13)	0.370	0.621	0.282	0.439	0.845	0.538	0.382	0.531	0.610	0.853	0.854	0.824	0.670	0.675	0.653	0.610
	ATSS (D14)	0.785	0.830	0.779	0.843	0.924	0.797	0.871	0.678	0.844	0.925	0.935	0.875	0.841	0.825	0.826	0.839
	RepPoints (D15)	0.326	0.491	0.240	0.459	0.854	0.429	0.300	0.349	0.534	0.874	0.864	0.720	0.658	0.679	0.458	0.549
APPA	YOLOv5-s (D1)	0.198	0.548	0.272	0.327	0.915	0.327	0.372	0.569	0.756	0.905	0.915	0.904	0.799	0.858	0.849	0.634
	YOLOv6-s (D2)	0.618	0.383	0.392	0.582	0.924	0.451	0.530	0.075	0.564	0.936	0.915	0.903	0.862	0.876	0.860	0.658
	YOLOv7-l (D3)	0.320	0.583	0.289	0.252	0.935	0.350	0.471	0.665	0.812	0.925	0.905	0.906	0.810	0.858	0.841	0.661
	YOLOv8-s (D4)	0.391	0.556	0.348	0.101	0.884	0.374	0.464	0.513	0.730	0.905	0.854	0.879	0.797	0.818	0.783	0.626
	RTMDet-t (D5)	0.746	0.817	0.673	0.734	0.883	0.747	0.820	0.701	0.828	0.884	0.904	0.900	0.820	0.845	0.826	0.809
	YOLOv5-n (D6)	0.365	0.447	0.238	0.353	0.905	0.258	0.395	0.153	0.621	0.915	0.875	0.901	0.821	0.857	0.837	0.596
	YOLOv6-t (D7)	0.490	0.556	0.478	0.404	0.944	0.435	0.336	0.670	0.713	0.906	0.916	0.917	0.851	0.879	0.891	0.692
	YOLOv7-t (D8)	0.667	0.598	0.498	0.683	0.924	0.573	0.642	0.157	0.644	0.925	0.894	0.894	0.841	0.865	0.837	0.709
	YOLOv8-n (D9)	0.327	0.363	0.209	0.140	0.864	0.346	0.314	0.204	0.435	0.893	0.824	0.780	0.703	0.738	0.651	0.519
	RTMDet-s (D10)	0.720	0.829	0.679	0.747	0.873	0.760	0.826	0.746	0.852	0.864	0.895	0.882	0.807	0.816	0.817	0.808
	Faster R-CNN (D11)	0.791	0.822	0.749	0.727	0.874	0.782	0.810	0.765	0.800	0.905	0.895	0.874	0.823	0.831	0.842	0.819
	Swin-t (D12)	0.783	0.844	0.759	0.809	0.874	0.784	0.848	0.739	0.832	0.884	0.885	0.868	0.800	0.746	0.794	0.817
	RetinaNet (D13)	0.544	0.699	0.472	0.655	0.905	0.610	0.662	0.632	0.772	0.875	0.895	0.882	0.751	0.788	0.805	0.730
	ATSS (D14)	0.673	0.797	0.641	0.729	0.874	0.703	0.763	0.692	0.788	0.875	0.885	0.840	0.728	0.663	0.724	0.758
	RepPoints (D15)	0.623	0.587	0.370	0.661	0.885	0.593	0.635	0.435	0.698	0.875	0.905	0.850	0.739	0.796	0.701	0.690
APPA-360	YOLOv5-s (D1)	0.113	0.428	0.193	0.214	0.925	0.249	0.288	0.584	0.726	0.864	0.915	0.904	0.790	0.839	0.850	0.592
	YOLOv6-s (D2)	0.383	0.299	0.260	0.323	0.893	0.265	0.482	0.208	0.555	0.945	0.884	0.882	0.831	0.856	0.837	0.594
	YOLOv7-l (D3)	0.224	0.491	0.154	0.173	0.934	0.261	0.351	0.645	0.775	0.916	0.905	0.904	0.798	0.838	0.829	0.613
	YOLOv8-s (D4)	0.337	0.448	0.230	0.079	0.874	0.287	0.345	0.462	0.679	0.915	0.854	0.877	0.785	0.817	0.750	0.583
	RTMDet-t (D5)	0.738	0.822	0.631	0.629	0.802	0.737	0.773	0.795	0.777	0.863	0.884	0.903	0.832	0.849	0.871	0.794
	YOLOv5-n (D6)	0.229	0.322	0.156	0.099	0.905	0.129	0.232	0.234	0.560	0.915	0.855	0.878	0.776	0.816	0.796	0.527
	YOLOv6-t (D7)	0.201	0.441	0.259	0.168	0.924	0.277	0.207	0.657	0.681	0.915	0.895	0.907	0.822	0.869	0.871	0.606
	YOLOv7-t (D8)	0.638	0.407	0.355	0.579	0.914	0.423	0.514	0.098	0.539	0.935	0.905	0.905	0.854	0.877	0.859	**0.653**
	YOLOv8-n (D9)	0.325	0.315	0.200	0.207	0.843	0.349	0.190	0.080	0.212	0.883	0.803	0.719	0.698	0.740	0.586	0.477
	RTMDet-s (D10)	0.464	0.735	0.473	0.472	0.832	0.578	0.641	0.575	0.715	0.811	0.823	0.805	0.707	0.687	0.743	0.671
	Faster R-CNN (D11)	0.792	0.813	0.746	0.726	0.865	0.786	0.811	0.764	0.800	0.915	0.895	0.864	0.813	0.821	0.831	0.816
	Swin-t (D12)	0.782	0.852	0.766	0.798	0.873	0.784	0.850	0.747	0.829	0.875	0.875	0.856	0.785	0.735	0.783	0.813
	RetinaNet (D13)	0.406	0.668	0.453	0.661	0.904	0.564	0.573	0.649	0.764	0.875	0.895	0.868	0.719	0.757	0.775	0.702
	ATSS (D14)	0.660	0.806	0.577	0.728	0.864	0.695	0.736	0.682	0.787	0.874	0.875	0.837	0.737	0.632	0.705	0.746
	RepPoints (D15)	0.385	0.624	0.329	0.630	0.885	0.532	0.545	0.517	0.729	0.875	0.885	0.811	0.709	0.702	0.628	0.652
DGA	YOLOv5-s (D1)	0.043	0.292	0.087	0.233	0.883	0.174	0.151	0.278	0.566	0.864	0.845	0.797	0.723	0.709	0.681	**0.488**
	YOLOv6-s (D2)	0.109	0.094	0.082	0.202	0.853	0.111	0.197	0.080	0.260	0.893	0.813	0.775	0.718	0.729	0.667	**0.439**
	YOLOv7-l (D3)	0.139	0.312	0.036	0.095	0.904	0.288	0.266	0.401	0.708	0.895	0.895	0.833	0.753	0.774	0.740	**0.536**
	YOLOv8-s (D4)	0.497	0.555	0.233	0.015	0.884	0.358	0.344	0.198	0.617	0.894	0.874	0.856	0.776	0.795	0.761	0.577
	RTMDet-t (D5)	0.586	0.762	0.484	0.459	0.750	0.632	0.660	0.693	0.655	0.822	0.833	0.858	0.754	0.766	0.763	0.698
	YOLOv5-n (D6)	0.101	0.240	0.103	0.132	0.883	0.079	0.134	0.138	0.496	0.884	0.853	0.808	0.742	0.750	0.728	**0.471**
	YOLOv6-t (D7)	0.091	0.277	0.117	0.098	0.874	0.191	0.065	0.324	0.514	0.863	0.843	0.832	0.731	0.753	0.716	**0.486**
	YOLOv7-t (D8)	0.638	0.420	0.325	0.623	0.924	0.500	0.550	0.036	0.510	0.925	0.904	0.912	0.872	0.886	0.868	0.660
	YOLOv8-n (D9)	0.208	0.157	0.114	0.206	0.832	0.172	0.083	0.026	0.085	0.863	0.782	0.349	0.449	0.571	0.396	0.353
	RTMDet-s (D10)	0.546	0.731	0.457	0.810	0.811	0.647	0.635	0.582	0.700	0.755	0.821	0.752	0.659	0.587	0.657	0.677
	Faster R-CNN (D11)	0.797	0.839	0.703	0.796	0.863	0.802	0.823	0.640	0.807	0.894	0.854	0.704	0.728	0.662	0.714	0.775
	Swin-t (D12)	0.724	0.818	0.716	0.767	0.844	0.751	0.811	0.705	0.787	0.863	0.854	0.820	0.761	0.683	0.749	0.777
	RetinaNet (D13)	0.288	0.474	0.179	0.393	0.844	0.361	0.295	0.284	0.570	0.844	0.813	0.537	0.583	0.530	0.457	**0.497**
	ATSS (D14)	0.199	0.459	0.177	0.504	0.853	0.327	0.322	0.396	0.621	0.864	0.808	0.587	0.593	0.393	0.471	**0.505**
	RepPoints (D15)	0.238	0.317	0.132	0.383	0.843	0.336	0.255	0.235	0.486	0.863	0.793	0.509	0.501	0.534	0.308	**0.449**



Fig. 9. Comparison of digital-level attack results using the real-world scenarios patch evaluation method.

Despite current remote sensing patch attacks exhibiting inferior performance on some models, the DGA method proposed in this study demonstrates a higher attack success rate in comparison with alternative approaches. Enhancing the generalization ability of patch attacks across various detectors is undoubtedly a valuable research direction, but that is not the

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TABLE V
SHOOTING DIRECTION ROBUSTNESS EVALUATION OF WHITE-BOX PHYSICAL-LEVEL ATTACKS

		180°	135°	90°	45°	0°	-45°	-90°	-135°	Avg.
YOLOv5-s	Clean	0.802	0.653	0.604	0.802	0.604	0.802	0.604	0.802	0.709
	OBJ	0.208	0.158	0.158	0.257	0.158	0.208	0.455	0.604	0.276
	APPA	0.257	0.208	0.109	0.347	0.109	0.604	0.208	0.307	0.269
	APPA-360	0.208	0.059	0.109	0.000	0.158	0.000	0.257	0.158	0.119
	DGA	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	**0.000**
YOLOv6-s	Clean	0.347	0.406	0.653	0.505	0.307	0.693	0.693	0.802	0.551
	OBJ	0.158	0.257	0.455	0.109	0.000	0.347	0.450	0.406	0.273
	APPA	0.208	0.307	0.455	0.406	0.059	0.406	0.632	0.941	0.427
	APPA-360	0.059	0.307	0.347	0.109	0.109	0.347	0.491	0.257	0.253
	DGA	0.059	0.059	0.109	0.000	0.000	0.158	0.000	0.000	**0.048**
YOLOv7-l	Clean	0.257	0.347	0.208	0.208	0.208	0.505	0.307	0.406	0.306
	OBJ	0.059	0.059	0.059	0.059	0.059	0.059	0.109	0.109	0.072
	APPA	0.059	0.208	0.059	0.208	0.059	0.208	0.208	0.347	0.170
	APPA-360	0.059	0.059	0.059	0.059	0.059	0.000	0.000	0.000	0.037
	DGA	0.000	0.059	0.000	0.000	0.000	0.000	0.000	0.000	**0.007**
YOLOv8-s	Clean	0.653	1.000	0.505	0.802	0.604	0.851	0.554	0.851	0.728
	OBJ	0.059	0.347	0.059	0.208	0.000	0.307	0.059	0.158	0.150
	APPA	0.406	0.653	0.257	0.505	0.307	0.554	0.406	0.653	0.468
	APPA-360	0.307	0.208	0.208	0.059	0.158	0.208	0.158	0.257	0.195
	DGA	0.000	0.208	0.059	0.059	0.109	0.257	0.000	0.059	**0.094**
RTMDet-t	Clean	1.000	0.941	0.901	1.000	1.000	0.941	0.941	1.000	0.966
	OBJ	0.941	1.000	0.752	0.901	0.802	1.000	0.851	1.000	0.906
	APPA	0.941	1.000	0.752	1.000	0.802	1.000	0.901	1.000	0.925
	APPA-360	0.851	0.851	0.505	0.752	0.653	0.941	0.653	0.941	**0.768**
	DGA	0.802	0.941	0.505	0.901	0.604	0.901	0.653	0.941	0.781

focus of this article. Our research is committed to improving the attack success rate in scenarios where the camera shooting direction is unknown, which significantly bolsters the practical applicability of patch attacks.

2) Comparison of Training Time: The DGA proposed in this article does not alter the number of model parameters compared to previous methods such as OBJ and APPA. The increase in computational load primarily stems from constructing the rotation matrix based on target orientation information and implementing the corresponding affine transformation. As demonstrated in Table III, we have compiled the training times of various patch attack algorithms and found that the training time for our method has only slightly increased. Given that the trained patches are printed and affixed to the targets and do not require real-time updates, this study's demand for rapid patch training is not critical.

3) Impact of Training Techniques: We observed that the RTMDet model exhibits significantly higher robustness against patch attacks compared to other models, prompting us to further investigate the disparities among different detectors. By analyzing the differences during the training stages of different detectors, we have identified two methods that can effectively enhance the robustness of patches.

As shown in Table IV, RTMDet-Tiny by default employs MixUp [43] data augmentation and the cosine annealing learning rate adjustment strategy [44], whereas detectors from the YOLO series, represented by YOLOv8, do not use these two training techniques. Upon the removal of these techniques, the detection performance of RTMDet against white-box attacks decreased from 0.750 to 0.313. Conversely, when these techniques were incorporated, the detection performance of YOLOv8 facing white-box attacks improved from 0.085 to 0.557. The experimental results indicate that these two training techniques greatly influence the detector's defensive capabilities against patch attacks, particularly the mixup data augmentation technique. Mixup blends two images in varying proportions, manifesting as one image superimposed on another. This is quite similar to the manifestation of patch attacks. Cosine annealing gradually reduces the learning rate following a cosine curve. This approach allows the model to initially learn quickly and then fine-tune the learned features as the learning rate decreases. This fine-tuning can lead the model to a more refined understanding of the data, potentially improving robustness against subtle adversarial perturbations. Therefore, introducing MixUp data augmentation and cosine annealing learning rate adjustment strategy during the training phase can enhance the model's robustness to patch attacks.

D. Physical-Level Attack

1) Overall Experimental Results: In this section, 1:43 proportionally scaled experiments are conducted to verify the physical level's attack performance. Specifically, we train five mainstream aerial detectors as victim models and record the AP of 20 vehicles under shooting direction robustness evaluation to compare the physical-level attack efficacy. By rotating the image around the center point as shown in Fig. 2, target photographs taken from eight different directions are obtained. The rotation process ensures that the size of the targets does not change.

First, we conducted white-box attack experiments on the selected five detectors and recorded the AP of eight shooting directions, as shown in Table V and Fig. 10. It is observed that the following conditions hold.

1) The proposed DGA attack has the smallest average AP at eight shooting directions in white-box attack experiments of four aerial detectors, far lower than the other three attack methods.

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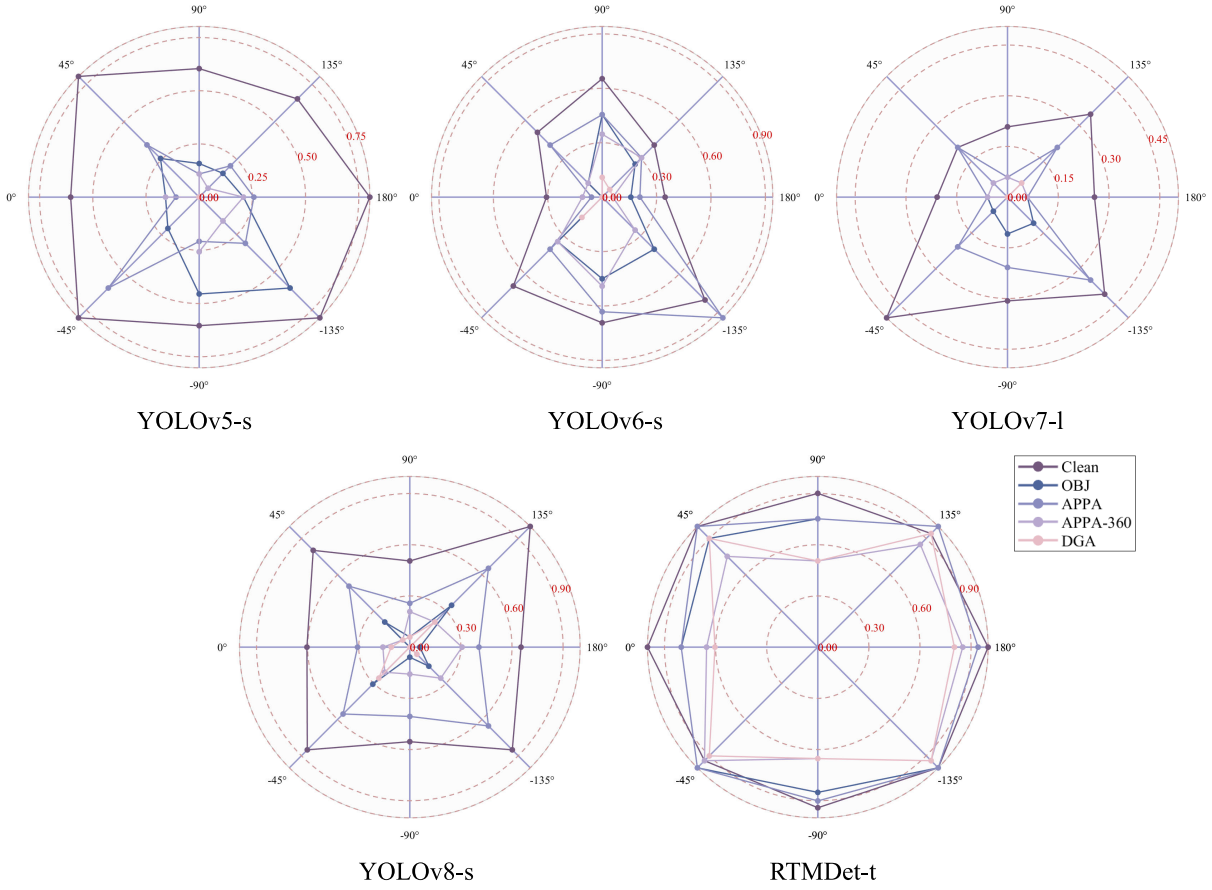


Fig. 10. Radar diagrams of white-box physical-level attacks.

2) Four attack methods do not perform well on the RTMDet-t detector, but APPA-360 and DGA performed slightly better than the other two methods.
3) Even though the detection performance of different detectors is different at each shooting direction, at the direction with larger AP in clean case, our method DGA attack usually has a relatively larger AP.
4) At vertical and horizontal directions, our DGA is easier to attack the detector.

Next, we supplemented the black-box attack experiments at the physical level and recorded the average values of eight shooting directions' AP, as shown in Fig. 11. Green represents the white-box attacks, black represents the black-box attacks, and bold represents optimal. It is observed that the following conditions hold.

1) The proposed DGA attack has the smallest average AP at eight shooting directions in black-box attack experiments of five aerial detectors, far lower than the other three attack methods.
2) The RTMDet-t detector shows robustness against various black- and white-box attacks, while the YOLOv7-l detector shows vulnerability. This is related to their detection performance of car models at the physical level.
3) In all the four attacks to the five detectors, the attack performance of the detector on which the attack is launched is not necessarily the best among all the detectors. In other words, the assumed detector does not determine the detector under the best attack effect. However, targeted attack works better in its corresponding detector.
4) Four attacks have better performance, especially in APPA and APPA-360, when the attack is launched on RTMDet-t. It is speculated that RTMDet-t is robust against the white- and black-box attacks, and the attack launched on it is more challenging and works better in other detectors.
5) The detectors with the strongest robustness against the four patch attack methods—OBJ, APPA, APPA-360, and DGA—were identified as YOLOv6-s, YOLOv6-s, YOLOv6-s, and YOLOv8-s, respectively. YOLOv6-s demonstrates strong resistance against various patch attacks at the physical level.

2) Detailed Experimental Results of CSDA Scenarios: We report the detailed quantitative and qualitative results in a CSDA setting, as shown in Table V and Figs. 12 and 13. It is observed that the following conditions hold.

1) When the image is clean, the detector produces better detection results for shooting directions of 45°, 135°, −45°, and −135° than for shooting directions of 0°, 90°, 180°, and −90°.
2) Models with colors white, black, and red are more easily detected, which means that they are more difficult to attack by patches. This may be related to the color distribution of vehicles in the training set.

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		D1	D2	D3	D4	D5	Avg.
Clean		0.709	0.551	0.306	0.728	0.966	0.652
OBJ	YOLOv5-s(D1)	0.276	0.290	0.070	0.412	0.949	0.399
	YOLOv6-s(D2)	0.637	0.273	0.114	0.564	0.936	0.505
	YOLOv7-l(D3)	0.479	0.329	0.072	0.552	0.980	0.482
	YOLOv8-s(D4)	0.208	0.114	0.043	0.150	0.880	0.279
	RTMDet-t(D5)	0.214	0.225	0.064	0.269	0.906	0.336
APPA	YOLOv5-s(D1)	0.269	0.385	0.127	0.448	0.942	0.434
	YOLOv6-s(D2)	0.612	0.427	0.250	0.764	0.993	0.609
	YOLOv7-l(D3)	0.323	0.361	0.170	0.510	0.980	0.469
	YOLOv8-s(D4)	0.504	0.345	0.140	0.468	0.948	0.481
	RTMDet-t(D5)	0.355	0.188	0.070	0.400	0.925	0.388
APPA-360	YOLOv5-s(D1)	0.119	0.213	0.051	0.274	0.968	0.325
	YOLOv6-s(D2)	0.425	0.253	0.075	0.515	0.941	0.442
	YOLOv7-l(D3)	0.255	0.268	0.037	0.403	0.936	0.380
	YOLOv8-s(D4)	0.324	0.160	0.036	0.195	0.887	0.320
	RTMDet-t(D5)	0.249	0.194	0.051	0.245	**0.768**	0.301
DGA	YOLOv5-s(D1)	**0.000**	0.080	0.007	0.131	0.816	**0.207**
	YOLOv6-s(D2)	0.054	**0.048**	0.030	0.137	0.800	**0.214**
	YOLOv7-l(D3)	0.043	0.092	**0.007**	0.138	0.930	**0.242**
	YOLOv8-s(D4)	0.208	0.068	0.022	**0.094**	0.879	**0.254**
	RTMDet-t(D5)	0.103	0.146	0.028	0.116	0.781	**0.235**



Fig. 11. Comparison of physical-level attack results.

3) At different attacks, detectors, and shooting directions, some specific models are easier to detect. This may be related to their relative location, color, and surroundings.
4) DGA does have the best attack effect in different shooting directions and on different detectors.
5) RTMDet-t demonstrates strong resistance against various patch attacks in CSDA scenarios.

3) Sensitivity Analysis of β: Parameter β is the weight of L_{tv}, which was initially intended to smooth the texture of the adversarial patches, thereby making them more easily detectable by cameras and enhancing the success rate of attacks in real-world scenarios. To substantiate this, we have conducted an extra set of physical-level experiments to assess the impact of parameter β on real-world effectiveness. Specifically, we maintained the α parameter constant at 0.01 and varied the β parameter to train and obtain five different patches. Subsequently, we printed ten slices for each type of patch and pasted them onto ten car models to photograph and test the average detection accuracy of the YOLOv5-s model at various angles. The patterns of the patches used and the experimental results are illustrated in Fig. 14. The results demonstrate that the optimal attack effect in the real world is achieved when the parameter α is set to 2.5. When the parameter α is set too low, the patch presents a mosaic effect where many details are lost and cannot be captured and preserved by the camera, leading to a significant discrepancy between physical- and digital-level attack efficacy. Conversely, when the parameter α is too high, the patch details may be well preserved by the camera, but the texture becomes overly simplistic and the colors tend to be monotonous, which severely diminishes the attack's effectiveness.

4) Ablation Study of Patch's Direction: We report the detailed quantitative and qualitative results in various patch direction settings, as shown in Table VI and Fig. 15, respectively. It is observed that the following conditions hold.

1) This experiment once again confirms that the direction of the patch does indeed have an impact on the effectiveness of the attack. When the patch placement direction is consistent with the training direction, the attack efficacy is best.
2) Models with black and white colors are more sensitive to changes in patch direction, which can easily lead to attack failure. This may be because the colorful patches have a higher contrast with black and white.

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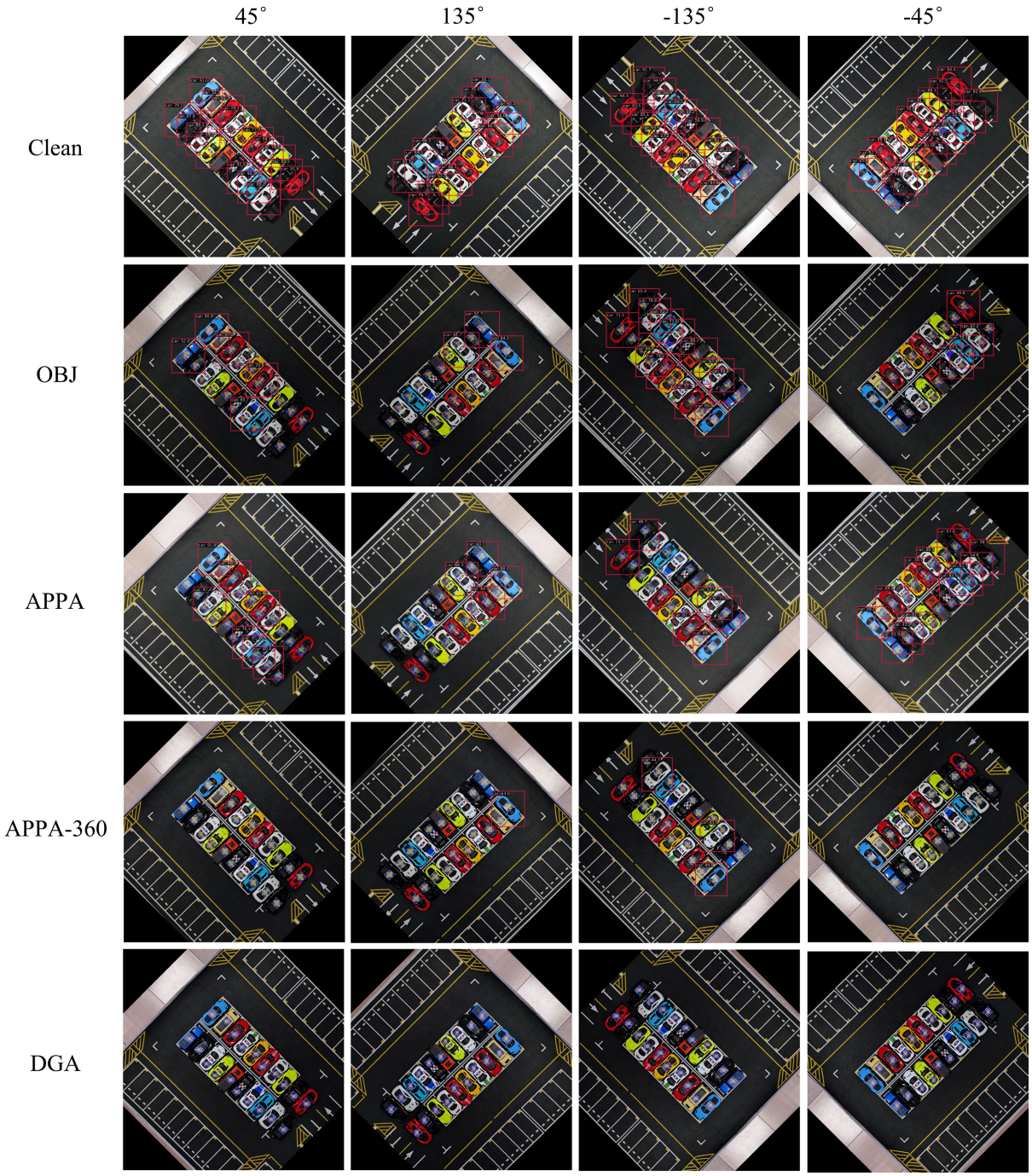


Fig. 12. YOLOv5-s' qualitative results of white-box physical-level attacks when shooting direction is 45°, 135°, 225°, and 315°.

TABLE VI

COMPARISON OF WHITE-BOX PHYSICAL-LEVEL ATTACK RESULTS WHEN THE PLACEMENT DIRECTION OF PATCHES CHANGES

	180°	135°	90°	45°	0°	-45°	-90°	-135°	Avg.
-90°	0.257	0.257	0.000	0.109	0.109	0.347	0.059	0.059	0.150
0°	0.000	0.208	0.059	0.059	0.109	0.257	0.000	0.059	**0.094**
90°	0.158	0.208	0.208	0.109	0.208	0.505	0.109	0.208	0.214
180°	0.208	0.257	0.000	0.158	0.000	0.307	0.059	0.158	0.143

3) Rotating the patches 90°and −90°, the attack effect declines more obviously.

5) Analysis of Aerial Detector's Attention: To further analyze the principle of the patch-based attack, we visualized the attention of YOLOv5-s obtained by GradCam [45] under different patch attacks, as shown in Fig. 16. It is observed that the following conditions hold.

1) From the heat map of the clean image, it can be seen that the aerial detector's attention to the vehicles is focused on the center point (roof) of the vehicle.

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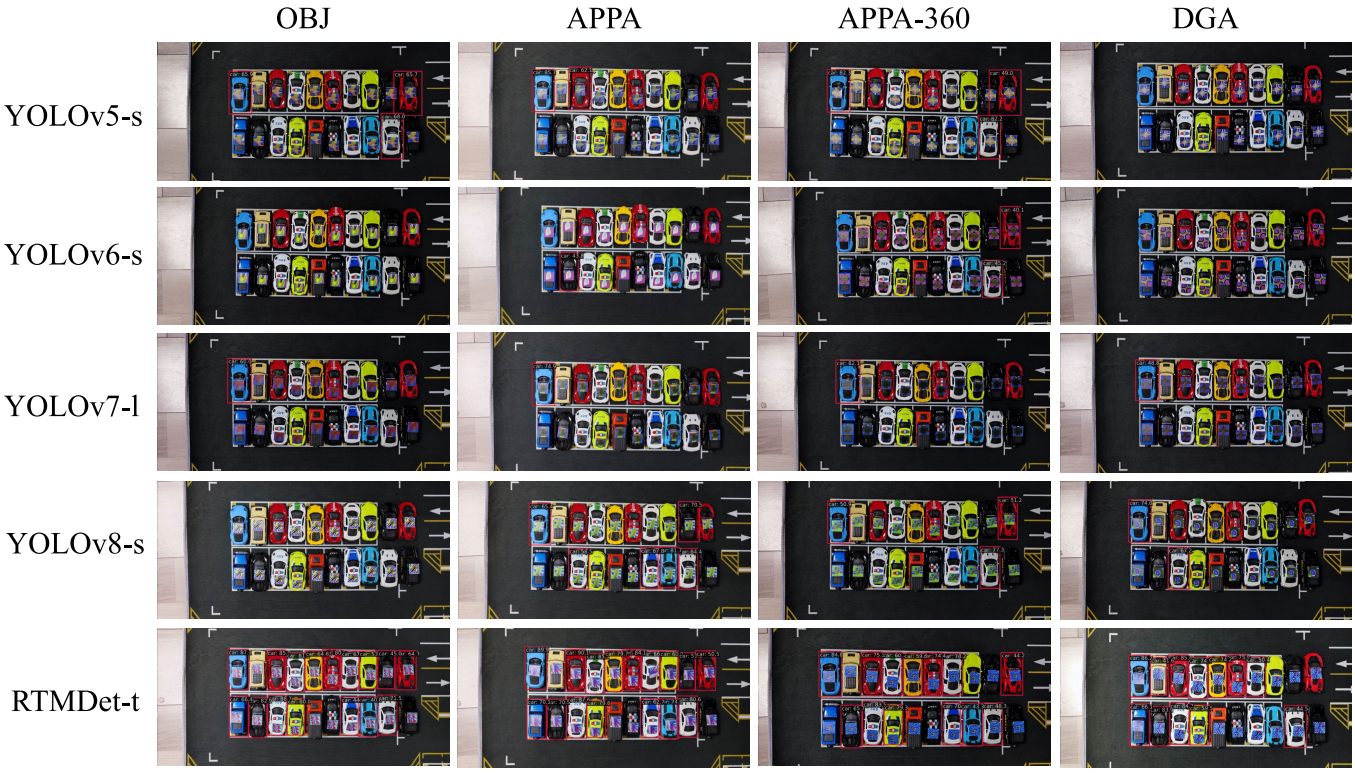


Fig. 13. Qualitative results of white-box physical-level attacks against five different real-time detectors.

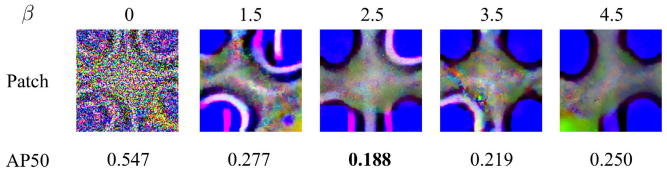


Fig. 14. Sensitivity analysis of β at physical level.

2) The patches that can disperse the previously focused areas (red regions) on the car models have better attack effectiveness.
3) Models with attention that is poorly dispersed are easier to be detected.
4) DGA disperses the attention at the roof of the vehicle more effectively.

6) Ablation Study of Patch's Position: Since the aerial detector's attention to the vehicles is focused on the roof of the vehicle, we design an ablation experiment where patches are installed at different positions on the vehicle. We report the detailed quantitative and qualitative results in three position settings, as shown in Table VII and Fig. 17, respectively. It is observed that the following conditions hold.

1) The patches installed on the vehicles' head disperse the attention of the aviation detector, which means that the patch can distract the original attention.
2) Although the patches installed on the vehicle's heads do not block the vehicle's roofs, they can still successfully attack, which indicates that blocking the key information of the target is not the main reason for the patch attack.

E. Real-World-Level Attack

To further demonstrate the superiority of the patch-based attack method proposed in this article, we conducted real-world-level attack experiments using real vehicles and large patches with the size of 1.2 × 1.2 m.

1) Overall Experimental Results: First, we selected five aerial detectors and printed their large adversarial patches generated through four methods: OBJ, APPA, APPA-360, and DGA. Next, we controlled the drone to take photographs of the target vehicle from heights of 20, 40, 60, 80, 100, and 120 m. Then, we recorded the average success rate of the four attack methods at six different heights. We report the detailed quantitative and qualitative results in three position settings, as shown in Table VIII and Fig. 18, respectively. It is observed that the following conditions hold.

1) The proposed DGA attack has the highest success rate in white-box attack experiments on five aerial detectors, far higher than the other three attack methods.
2) The RTMDet-t detector has high robustness against real-world-level patch attacks, and even the most effective DGA attack only has a success rate of 10.4%.
3) In real-world-level attack experiments, the highest success rate of attacks is only 77.1%, which is not yet 100%.
4) From Fig. 18, it can be seen that DGA exhibits stronger robustness against multiscale targets compared to the other three attack methods.

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Fig. 15. YOLOv8-s' qualitative results of white-box physical-level attacks when the placement direction of patches changes.

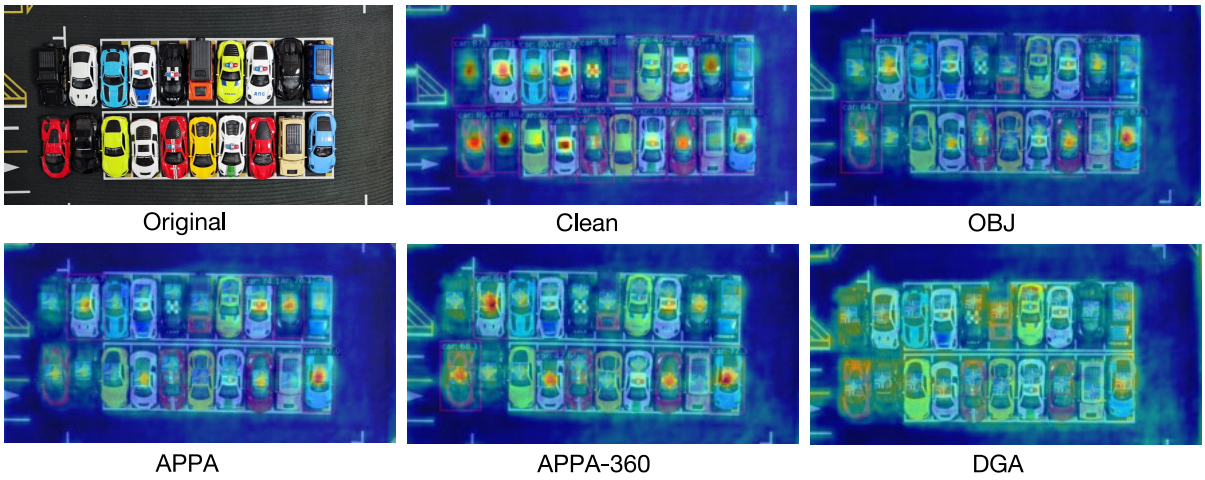


Fig. 16. Visualization of the YOLOv5-s' attention before and after different patch-based attacks at the physical level.

TABLE VII
COMPARISON OF YOLOV8-S' PHYSICAL-LEVEL ATTACK RESULTS WHEN THE PLACEMENT POSITION OF PATCHES CHANGES

	180°	135°	90°	45°	0°	-45°	-90°	-135°	Avg.
Head	0.000	0.376	0.000	0.129	0.129	0.257	0.000	0.505	**0.175**
Roof	0.257	0.376	0.129	0.129	0.376	0.624	0.000	0.129	0.253
Tail	0.376	0.129	0.129	0.129	0.376	0.624	0.257	0.624	0.331



Fig. 17. YOLOv8-s' qualitative results of white-box physical-level attacks when the placement position of patches changes.

5) There is no positive or negative correlation between the success rate of patch attacks and the shooting height. A higher shooting height does not necessarily mean a lower success rate of attacks.
6) The detection results between vehicles are not completely isolated, and a vehicle carrying the patch can also have some impact on the model's prediction results of surrounding vehicles.
7) RTMDet-t demonstrates strong resistance against various patch attacks at the real-world level.

2) Ablation Study of Lighting Conditions: Complex lighting conditions are an inescapable variable in the real-world deployment of adversarial patches. To address this, we conducted controlled experiments to investigate the performance of our patches during dusk, a time when lighting is inherently suboptimal. The outcomes of these experiments are illustrated in Fig. 19. In comparison to the well-lit conditions depicted in Fig. 18, the efficacy of the patches at dusk was significantly reduced. Even the most effective DGA method experienced a drastic drop, with only a 33.3% success rate in adversarial attacks. This finding underscores the importance of lighting conditions as a determinant of success in real-world adversarial patch applications, a factor not yet adequately accounted for in the design of current physical patch attack methodologies. Future work should incorporate variations in lighting conditions into the patch training framework, enhancing the robustness of patches against different lighting environments and furthering their practical applicability.

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TABLE VIII
COMPARISON OF ATTACK SUCCESS RATES FOR WHITE-BOX ATTACKS AT REAL-WORLD LEVEL

		180°	135°	90°	45°	0°	-45°	-90°	-135°	Avg.
YOLOv5-s	OBJ	33.3	16.7	66.7	33.3	100.0	83.3	33.3	66.7	54.2
	APPA	33.3	50.0	50.0	33.3	33.3	66.7	16.7	33.3	39.6
	APPA-360	16.7	33.3	33.3	66.7	50.0	33.3	16.7	66.7	39.6
	DGA	100.0	33.3	100.0	33.3	83.3	83.3	66.7	50.0	**68.7**
YOLOv6-s	OBJ	0.0	16.7	0.0	66.7	33.3	16.7	0.0	33.3	20.8
	APPA	16.7	0.0	0.0	66.7	33.3	0.0	0.0	16.7	16.7
	APPA-360	0.0	16.7	0.0	33.3	16.7	16.7	0.0	0.0	10.4
	DGA	33.3	33.3	0.0	83.3	83.3	33.3	33.3	16.7	**39.6**
YOLOv7-l	OBJ	66.7	66.7	16.7	33.3	66.7	50.0	33.3	66.7	50.0
	APPA	83.3	66.7	0.0	0.0	16.7	33.3	33.3	50.0	35.4
	APPA-360	33.3	50.0	16.7	33.3	50.0	16.7	16.7	16.7	29.2
	DGA	66.7	66.7	33.3	50.0	83.3	66.7	50.0	66.7	**60.4**
YOLOv8-s	OBJ	16.7	66.7	16.7	66.7	83.3	83.3	0.0	33.3	45.8
	APPA	50.0	83.3	0.0	66.7	83.3	66.7	0.0	16.7	45.8
	APPA-360	33.3	33.3	0.0	33.3	83.3	50.0	0.0	33.3	33.3
	DGA	100.0	100.0	83.3	66.7	100.0	83.3	16.7	66.7	**77.1**
RTMDet-t	OBJ	0.0	16.7	0.0	0.0	0.0	0.0	0.0	0.0	2.1
	APPA	0.0	0.0	0.0	16.7	0.0	0.0	0.0	0.0	2.1
	APPA-360	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	DGA	0.0	50.0	0.0	16.7	16.7	0.0	0.0	0.0	**10.4**

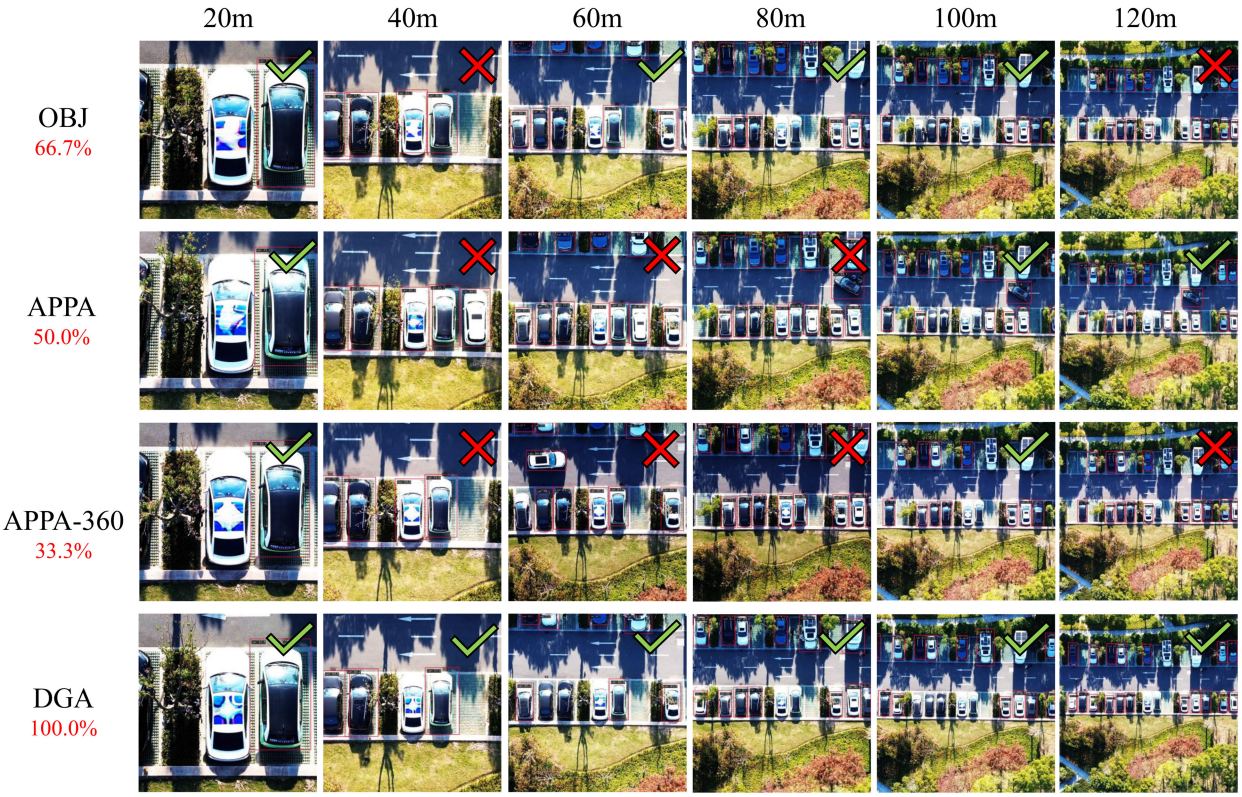


Fig. 18. YOLOv5-s' qualitative results of white-box real-world-level attacks when the camera shooting height changes.

3) Ablation Study of Adversarial Medium: We controlled the drone to take photographs of the target vehicle from pitch angles of 90°, 75°, 60°, 45°, and 30°. Then, we recorded the average success rate of the DGA attack at five different pitch angles. We report the detailed qualitative results in two adversarial medium settings, as shown in Fig. 20. It is observed that the following conditions hold.

1) The material of the adversarial medium can have an impact on real-world attack results.
2) The smaller the pitch angle of the camera, the greater the difficulty of the attack.
3) Since the coarse fibers and density of the textiles, when the pitch angle is small, the color of the patch is distorted.
4) Printing patches on synthesis paper has a better attack effect than printing them on textiles.

V. DISCUSSION

Although adversarial patches can fool aerial detectors, the presence of such patches is often easily uncovered by the naked eye. Therefore, if we want these patches to replace

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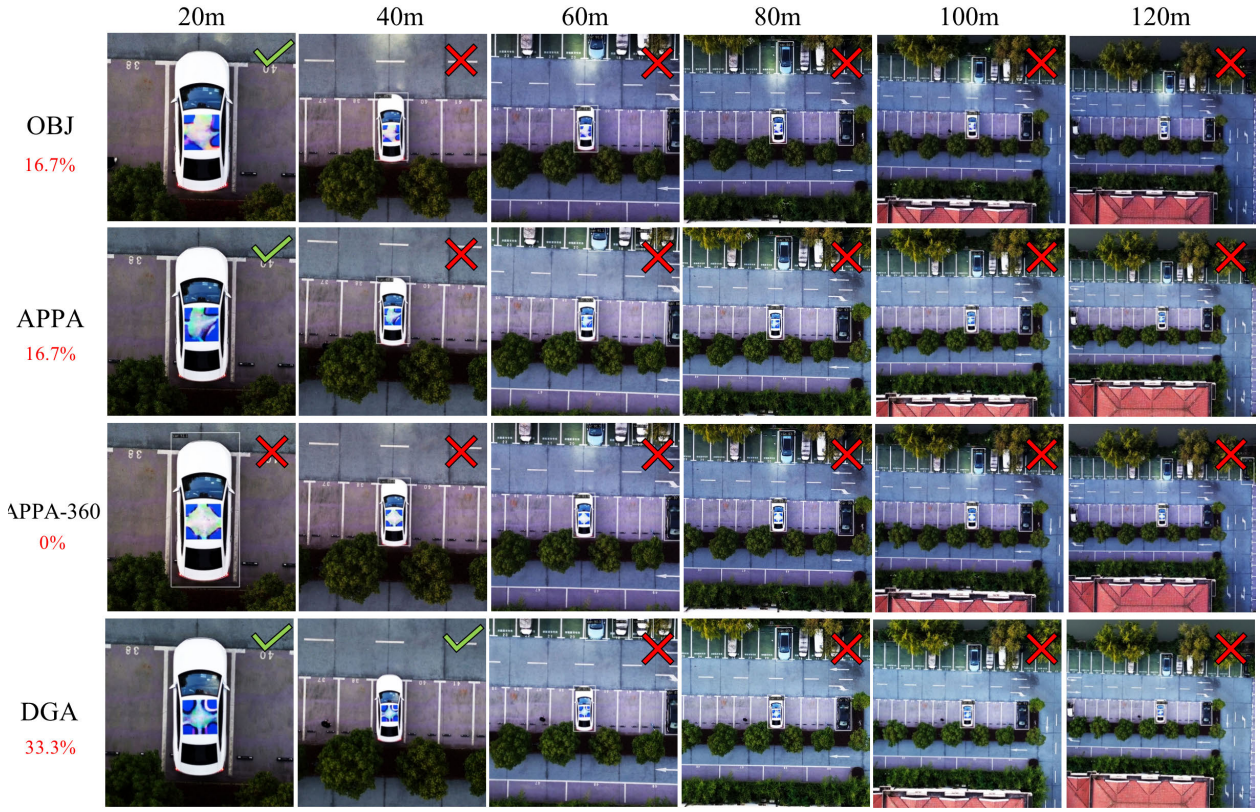


Fig. 19. YOLOv5-s' qualitative results of white-box real-world-level attacks when the camera shooting height changes during dusk with insufficient lighting.

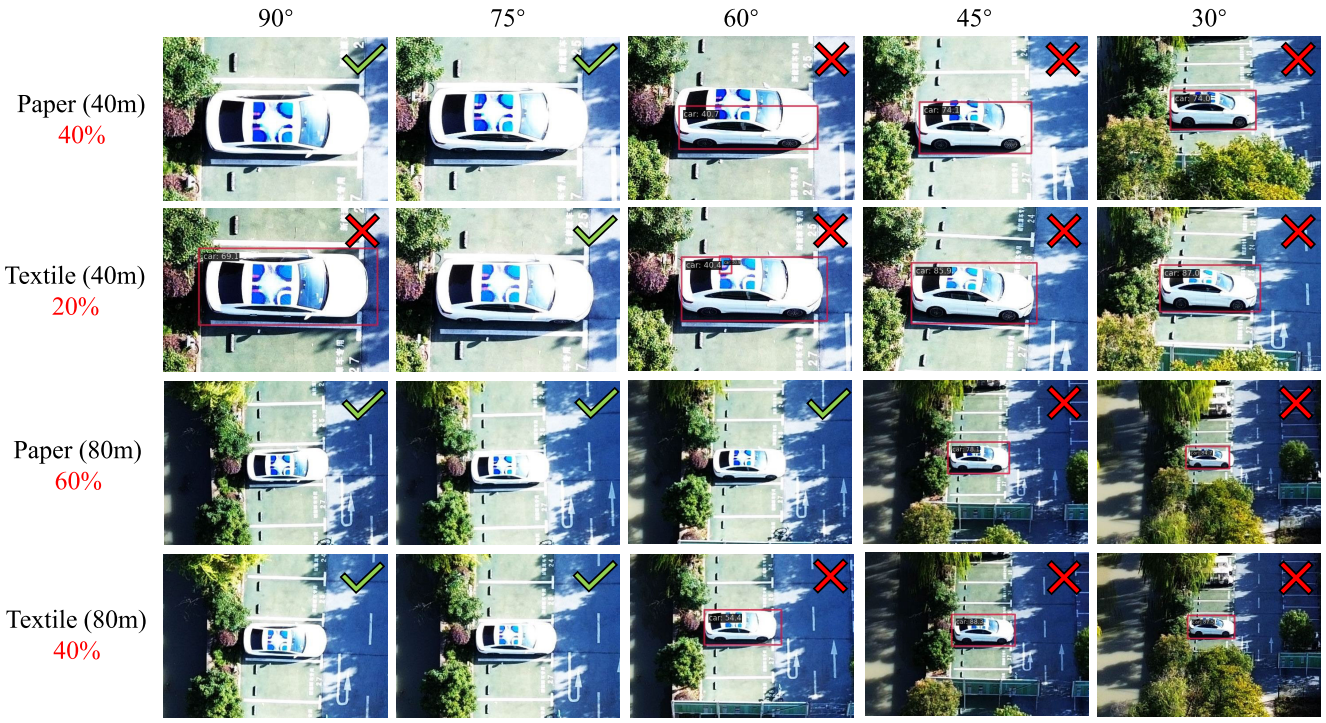
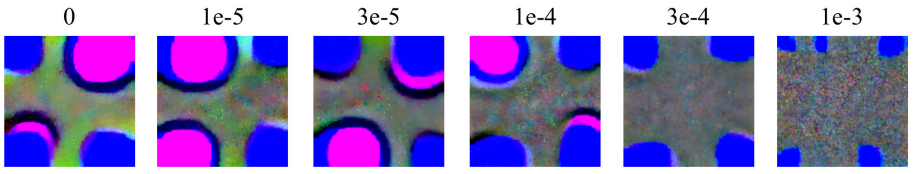


Fig. 20. YOLOv5-s' qualitative results of white-box real-world-level attacks when the material and pitch angle of patch change.

traditional camouflage, we must enhance their imperceptibility. Van Etten [46] attempts to modify the transparency of the patch to enhance its imperceptible performance, but at the cost of sacrificing the success rate of the attack. In this section, we used PerC [47] and PSNR [48] to evaluate the concealment of patches generated by different attack methods. As shown in Table IX, APPA has the lowest PerC distance, which means that it is relatively the least noticeable among the four attack

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PSNR	41.900	41.860	42.062	42.184	42.813	**43.336**
PerC	2336.088	2292.111	2193.972	2109.319	1862.053	**1738.432**
AP50	**0.079**	0.091	0.095	0.131	0.222	0.388

Fig. 21. Sensitivity experiment of weight on perceptual loss function.

TABLE IX
EVALUATION OF YOLOV5-S PATCH'S IMPERCEPTIBILITY

	PerC ↓	PSNR ↑
OBJ	2192.01	**42.45**
APPA	**2181.45**	42.33
APPA-360	2185.08	42.42
DGA	2225.69	42.17

methods. DGA has the highest PerC distance and PSNR, which means that it is relatively the easiest to detect by the naked eye among the four attack methods.

In scenarios where manual detection can be involved, the stealthiness of adversarial patches is paramount. A more stringent quantitative analysis was undertaken to evaluate the balance between the effectiveness of attacks and their ability to evade human detection. Following the approach outlined in [30], we incorporated a perceptual loss function to improve the stealthiness of the patches. The results of these experiments are depicted in Fig. 21. The first row presents the patches generated with varying weights of the PerC loss function, while the subsequent three rows display three metrics corresponding to each patch. As the weight of the perceptual loss function increases, the PerC distance decreases, and the PSNR increases, indicating better visual quality. However, there is a noticeable decline in attack performance. It can be observed from Fig. 21 that colors with high saturation levels, which are easily detectable by the human eye (e.g., pink), gradually diminish as the loss of weight increases. Conversely, colors to which the human eye is less sensitive, such as shades of gray, become more prevalent.

This experiment reveals an inherent tradeoff between the stealth of the patch and its attack capability. It is a balancing act between the two, where improvements in one aspect may lead to compromises in the other. In the next work, we will focus on improving the imperceptibility of patches without affecting the attack efficacy.

VI. CONCLUSION

In this article, a brand-new direction-guided patch attack framework, DGA, against aerial detection is proposed for CSDA scenarios. We have corrected the incorrect patch-pasting method in the field of aerial detection in the past and propose a new real-world scenarios patch evaluation method. It can enable patches trained in the digital to have better attack effects in the real world. Extensive and rigorous experiments have been conducted on our SJTU-4K dataset to validate the proposed patch-based attack framework's effectiveness in CSDA scenarios. The white- and black-box experimental results at the levels of digital, physical, and real world have all verified the superiority of DGA. In addition, DGA shows powerful robustness to multiresolution aerial images.

APPENDIX

The complete derivation processes for the matrices $\boldsymbol{Mr}$, $\boldsymbol{Me}$, $\boldsymbol{Mi}$, and $\boldsymbol{R}$ are provided.

To simplify the analysis, we assume that the world coordinate system is consistent with the model coordinate system. First, the world coordinates (X_w, Y_w, Z_w) are transformed into camera coordinates (X_c, Y_c, Z_c) through the extrinsic matrix $\boldsymbol{M}_e$ as follows:

$$\begin{bmatrix} X_{\mathrm{c}} \\ Y_{\mathrm{c}} \\ Z_{\mathrm{c}} \\ 1 \end{bmatrix} = \boldsymbol{M}_e \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{M}_r & \boldsymbol{t} \\ \boldsymbol{0}^{\mathrm{T}} & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \tag{19}$$

where $\boldsymbol{t}$ is a 3×1 offset translation vector, $\boldsymbol{0}^{\mathrm{T}}$ is a 1×3 zero vector, and $\boldsymbol{M}_r$ is a 3×3 rotation matrix. Assuming that the camera coordinate system rotates by φ, α, and θ degrees around the x-, y-, and z-axes, respectively, the rotation matrix can be represented in (20), as shown at the top of the next page.

Assuming that t_x, t_y, and t_z represent the offsets of the optical center C relative to the origin of the model coordinate system along the x, y, and z axes, respectively, the extrinsic matrix $\boldsymbol{M}_e$ can be represented in (21), as shown at the top of the next page.

The transformation from the world coordinate system to the camera coordinate system is a conversion between 3-D coordinates, whereas the transformation from the camera coordinate system to the image coordinate system is a conversion from 3-D to 2-D coordinates. Assuming that the distance from the image plane to the optical center C is f, then we have

$$Z_c \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} \tag{22}$$

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$$\begin{aligned}\boldsymbol{M}_r &= \begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & \cos\varphi & \sin\varphi\\ 0 & -\sin\varphi & \cos\varphi\end{bmatrix}\begin{bmatrix}\cos\alpha & 0 & -\sin\alpha\\ 0 & 1 & 0\\ \sin\alpha & 0 & \cos\alpha\end{bmatrix}\\ &= \begin{bmatrix}\cos\theta\cos\alpha-\sin\theta\sin\varphi\sin\alpha & -\sin\theta\cos\varphi & -\cos\theta\sin\alpha-\sin\theta\sin\varphi\cos\alpha\\ \sin\theta\cos\alpha+\cos\theta\sin\varphi\sin\alpha & \cos\theta\cos\varphi & -\sin\theta\sin\alpha+\cos\theta\sin\varphi\cos\alpha\\ \cos\varphi\sin\alpha & -\sin\varphi & \cos\varphi\cos\alpha\end{bmatrix}\end{aligned} \tag{20}$$

$$\boldsymbol{M}_e = \begin{bmatrix}\cos\theta\cos\alpha-\sin\theta\sin\varphi\sin\alpha & -\sin\theta\cos\varphi & -\cos\theta\sin\alpha-\sin\theta\sin\varphi\cos\alpha & t_x\\ \sin\theta\cos\alpha+\cos\theta\sin\varphi\sin\alpha & \cos\theta\cos\varphi & -\sin\theta\sin\alpha+\cos\theta\sin\varphi\cos\alpha & t_y\\ \cos\varphi\sin\alpha & -\sin\varphi & \cos\varphi\cos\alpha & t_z\\ 0 & 0 & 0 & 1\end{bmatrix} \tag{21}$$

where (x, y) denotes the physical coordinates in the image coordinate system, with units in millimeters. To obtain the image pixel coordinate system (u, v), the following transformation is necessary:

$$\begin{bmatrix}u\\ v\\ 1\end{bmatrix} = \begin{bmatrix}\frac{1}{dx} & 0 & u_0\\ 0 & \frac{1}{dy} & v_0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ 1\end{bmatrix} \tag{23}$$

where dx and dy represent the number of millimeters per pixel in each column and row, respectively. u_0 and v_0 denote the origin offsets, with the origin of the image coordinate system typically located at the image's top-left corner. Also, we can employ the intrinsic matrix $\boldsymbol{M}_i$ to depict the transformation from the camera coordinate system to the image coordinate system, as illustrated in the following:

$$\begin{bmatrix}u\\ v\\ 1\end{bmatrix} = \boldsymbol{M}_i\begin{bmatrix}X_c\\ Y_c\\ Z_c\\ 1\end{bmatrix}. \tag{24}$$

Consequently, the intrinsic matrix is formulated as follows:

$$\boldsymbol{M}_i = \begin{bmatrix}\frac{f}{dx} & 0 & u_0 & 0\\ 0 & \frac{f}{dy} & v_0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}. \tag{25}$$

Ultimately, the matrix $\boldsymbol{R}$ can be expressed as follows:

$$\boldsymbol{R} = \boldsymbol{M}_i\boldsymbol{M}_e \tag{26}$$

where $\boldsymbol{M}_i$ denotes the intrinsic matrix and $\boldsymbol{M}_e$ denotes the extrinsic matrix. The camera's intrinsic matrix is generally invariant, while the extrinsic matrix changes with the variations in the camera's attitude.

When the camera captures images perpendicularly to the ground, the rotation angles of the camera's x-axis and y-axis are both 0. Consequently, $\boldsymbol{R}$ can be further simplified to

$$\begin{aligned}\boldsymbol{R} &= \begin{bmatrix}\frac{f}{dx} & 0 & u_0 & 0\\ 0 & \frac{f}{dy} & v_0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}\begin{bmatrix}\cos\theta & -\sin\theta & 0 & t_x\\ \sin\theta & \cos\theta & 0 & t_y\\ 0 & 0 & 1 & t_z\\ 0 & 0 & 0 & 1\end{bmatrix}\\ &= \begin{bmatrix}\frac{f\cos\theta}{dx} & -\frac{f\sin\theta}{dx} & u_0 & \frac{ft_x}{dx}+u_0t_z\\ \frac{f\sin\theta}{dy} & \frac{f\cos\theta}{dy} & v_0 & \frac{ft_y}{dy}+v_0t_z\\ 0 & 0 & 1 & t_z\end{bmatrix}\end{aligned} \tag{27}$$

where θ represents the angle of rotation about the z-axis of the camera coordinate system, as illustrated in Fig. 3.

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